



USER GUIDE



Rel.	Date	Description
0.0	30/04/2020	User Guide for Smartris Drive - first release
1.0	07/09/2020	Upgrade for new FW with improvements for Encoder Feedback
1.1	30/09/2020	Upgrade for new FW 1.0.7
1.2	20/10/2020	Upgrade for new FW 1.0.8
1.3	29/10/2020	Errata corride Upgrade for new FW 1.0.8



Release 1.3
Date 29/10/2020
This User Guide refers to FW 1.0.8

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REFERENCE DOCUMENTS:

- *Lafert CANOpen Manual rel. 1.2*
- *CiA 301 (310_1v01010005_cor.pdf)*
- *CiA 402 (CiA® 402 Draft Standard Proposal.pdf)*

1 | PURPOSE OF MANUAL

1.1 Purpose of the Manual

This operating guide provides information for safe installation and commissioning of the Smartris Drive.

Read and follow the instructions to use the Lafert Smartris Drive safely and professionally and pay attention to the safety instructions and general warnings.

Always keep this operating guide available with the Smartris Drive.

This operating guide provides information for safe installation and commissioning of the Lafert Smartris Drive: read carefully the entire guide before installing and using the equipment.

	Caution
	The operating guide is intended for use by qualified personnel.

A qualified person has the knowledge and authorization to perform tasks such as transporting, assembling, installing, commissioning and operating motors.

This manual is regularly reviewed and updated. All suggestions for improvement are welcome.

1.2 Warning Safety Information

In order to achieve the optimum, safe operation of the Smartris Drive, it is imperative that you implement the safety procedures included in this installation guide. This information is provided to protect you and to keep your work area safe when operating the Smartris Drive and accompanying equipment.

Safety Instructions: for the electrical installation, the ESD instructions must be observed.

	Caution
	• The Systems that are electrically connected must be properly secured so they cannot be switched back on and warnings signs must be put up. • Before startup, it must be checked that the wiring is correct and is free of mechanical damages. Only drive with wiring in perfect condition may be enabled to operation. • Incorrect voltage, reverse polarity and defective wiring can damage the drive. • Do not connect or disconnect electric cables while the equipment is powered or running. • The operator is responsible for keeping the safety installations in perfect working order, conforming to prevailing laws and standards.

Please read this chapter carefully before you begin the installation process.

The Lafert Smartris Drive contains electrostatic-sensitive components that can be damaged if handled incorrectly. To prevent any electrostatic damage, avoid contact with highly insulating materials, such as plastic film and synthetic fabrics. Place the product on a conductive surface and ground yourself in order to discharge any possible static electricity build-up.

To avoid any potential hazards that may cause severe personal injury or damage to the product during operation, keep all covers and cabinet doors shut.

The following safety symbols are used in this manual:

	Warning This information is needed to avoid a safety hazard, which might cause bodily injury or death as a result of incorrect operation: • To avoid electric arcing and hazards to personnel and electrical contacts, never connect/disconnect the servo drive while the power source is on. • Power cables can carry a high voltage, even when the motor is not in motion. Disconnect the Lafert Smartris Drive from all voltage sources before servicing. • After shutting off the power and removing the power source from your equipment, wait at least 1 minute before touching or disconnecting parts of the equipment that are normally loaded with electrical charges (such as capacitors or contacts). Measuring the electrical contact points with a meter, before touching the equipment, is recommended.
	Hot Surface Warning To alert against surfaces that may reach high temperatures. The heatsink and wires may reach high temperatures.
	Caution This information is necessary to prevent bodily injury, damage to the product or to other equipment: • The maximum DC power supply connected to the instrument must comply with the parameters outlined in this guide. • When connecting the Lafert Smartris Drive to an approved control supply, connect it through a line that is separated from hazardous live voltages using reinforced or double insulation in accordance with approved safety standards. • Before switching on the Smartris Drive, verify that all safety precautions have been observed and that the installation procedures in this manual have been followed. • Make sure that the Safe Torque Off is operational. • If a fire breaks out, do not direct the water extinguishers near the equipment to put out the flames.
	Important Identifies information that is critical for successful application and understanding of the product.

Safety measures must be taken both for people and machines, in compliance with Standards and local conditions.

1.3 Warranty Information

The products covered in this manual are warranted to be free of defects in material and workmanship and conform to the specifications stated either within this document or in the product catalogue description.

Do not remove any cover and warranty seal.

Interventions or modifications made to the equipment, or parts of it, and/to the standard fittings, which are not authorized by us, will void the warranty.

Any faults caused by failure to comply with instructions will NOT be covered by the warranty.

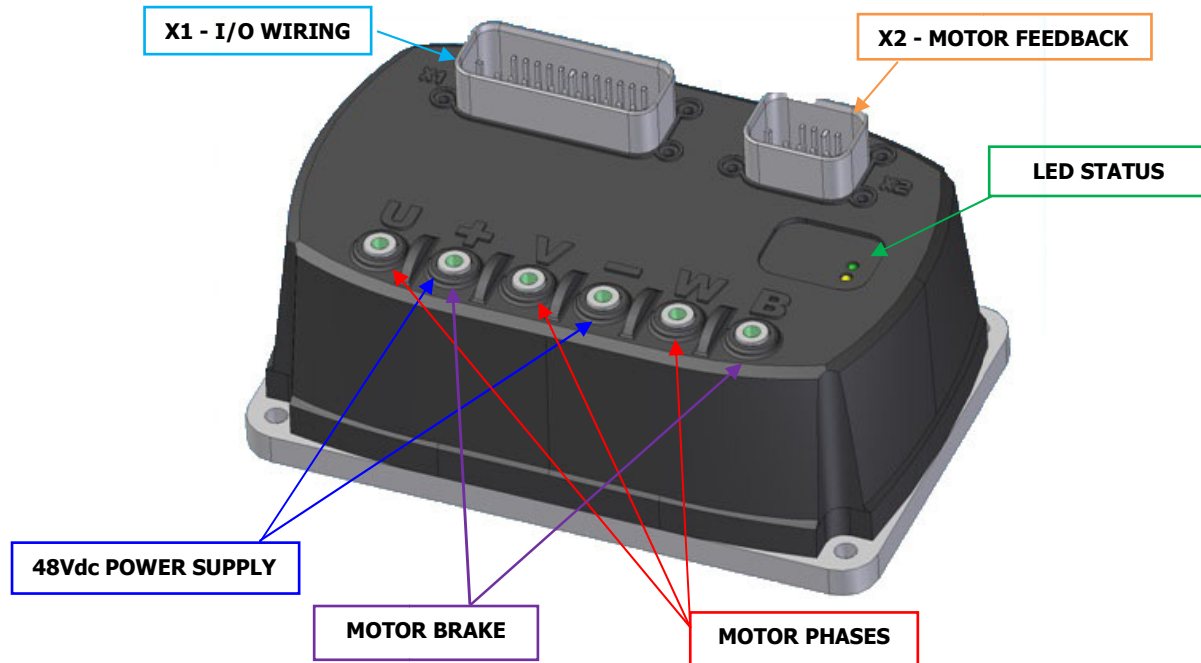
1.4 Electrical Specifications

Main power supply	
Supply Voltage	Typical 48 Vdc (30 Vdc Min/60Vdc Max)
STO Supply	24 Vdc (Dedicated Connection)

*for supply 24Vdc contact the manufacturer

Do not connect power supply voltages if higher than the admissible voltage range. If excessive voltages are applied to the drive, its internal components will be damaged.

1.5 Electrical Installation, Terminal Wirings



Picture 1 - Smartris Wiring

POWER SUPPLY SMARTRIS WIRING		
PIN	SIGNAL	DESCRIPTION
+	+ VBUS	Connection to + Vbus (+48 Vdc)
-	- VBUS	Connection to - Vbus

Table 1 - Power Supply Wiring

MOTOR PHASES SMARTRIS WIRING		
PIN	SIGNAL	DESCRIPTION
U	PHASE U	Motor Phase U
V	PHASE V	Motor Phase V
W	PHASE W	Motor Phase W

Table 2 - Motor Phases Wiring

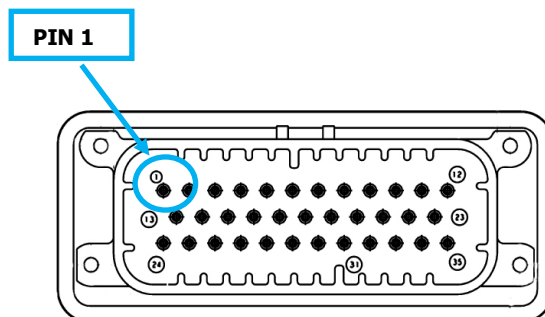
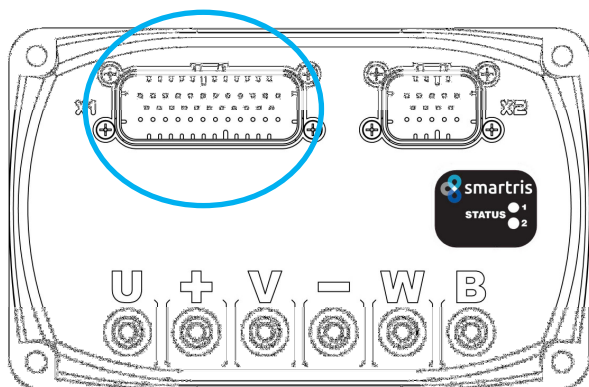
Recommended Cable: see chapter 3
Philip head screw M6, Tightening torques = 7 Nm

BRAKE		
PIN	SIGNAL	DESCRIPTION
+	+ VBUS	Motor Brake + (connect to + Vbus)
B	BRAKE	Motor Brake -

Table 3 - Brake Wiring

Minimum Recommended Cable: 1 mmq. -AWG19

X1 - INPUT/OUTPUT Connection



X1 - I/O SIGNAL			
PIN	SIGNAL	DESCRIPTION	COLOR
1	0V24	GND Control Board supply	Black
2	COMMON	Common Input	Pink
3	IN1	Digital Input 1	White/Pink
4	IN2	Digital Input 2	Yellow/Pink
5	IN3	Digital Input 3	Green/Pink
6	IN4	Digital Input 4	Brown/Pink
7	TX/485+	Tx232 Signal* / 485+ Signal	Gray/Brown
8	RX/485-	Rx232 Signal* / 485- Signal	Gray/Green
9	485_T	120 Ω Termination resistance 485 (connect to 485+)	White/Gray
10	GND_COM	GND RS232 Serial Communication	Gray
11	RET1	Safety Relays 1	White/Red
12	STO1	Safety Relays STO1	Yellow/Red
13	GND_REF	GND Analog Reference	Brown/Green
14	REF -	Analog Input Neg	Brown
15	REF +	Analog Input Pos	Yellow
16	+10V	10 Vdc Output	Violet
17	OUTDAC	Analog Output	White/Yellow

* RS232 protocol is used for service and drive configuration: for more information contact the manufacturer

X1 - I/O SIGNAL			
PIN	SIGNAL	DESCRIPTION	COLOR
18	GND_DAC	GND reference for Analog Output	White/Black
19	GND_CAN	GND reference for CAN	Brown/Black
20	CAN_T	120 Ω Termination resistance CAN (connect to CAN H)	White/Green
21	N.C	-	-
22	COM_RET	Common Relays	Brown/Red
23	STO2	Safety Relays STO2	Yellow/Brown
24	48V	Switching Supply	Red
25	V_OUT	Supply Output	Blue
26	OUT1	Digital Output 1 – DRIVE OK	White/Blue
27	OUT2	Digital Output 2 - Open Drain - Programmable	Gray/Blue
28	OUT3	Digital Output 3 - Open Drain - Programmable	Green/Blue
29	OUT4	Digital Output 4 – Brake Command (when brake is externally supplied)	Yellow/Blue
30	CAN_L	CAN_L Connection	Green
31	CAN_H	CAN_H Connection	White
32/33	N.C	-	-
34	RET2	Safety Relays 2	Green/Red
35	STO_COM	Common STO1 / STO2	Yellow/Black

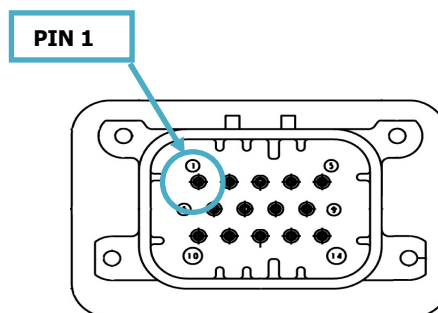
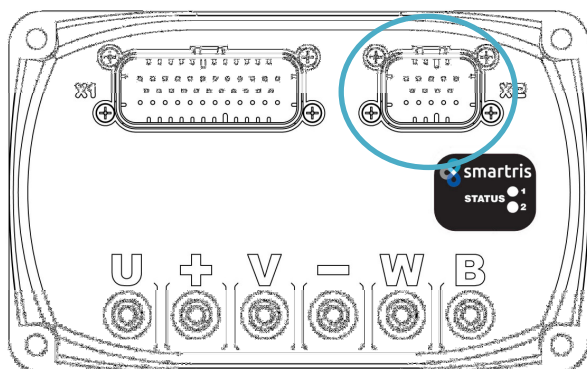
Table 4 - I/O Wiring

Recommended Cable cod. TE.CO. 5760 LIYCY 37x0,25

Diameter = 11,6 mm - Minimum bending radius = 6x $\varnothing$ cable = 70mm

DIGITAL INPUT/OUTPUT	
Impedance	3 KOhm, 7 mA
Digital Input	24 V +/- 20%
Digital Output	24 V +/- 20%

X2 - FEEDBACK



X2 - FEEDBACK– See Annex A to see Lafert Motor Wirings			
PIN	RESOLVER	INC ENC+HALL	SINCOS ENCODER
1	S2 (SIN+)	A+	SIN
6	S4 (SIN-)	A-	REFSIN
2	S1 (COS+)	B+	COS
7	S3 (COS-)	B-	REFCOS
3	R1 (REF+)	Z+	DATA+
8	R2 (REF-)	Z-	DATA-
10	-	+5V	+8V
11	-	GND	GND
4	-	HX	-
9	-	XY	-
14	-	HZ	-
5	SCH	SCH	SCH
12	NTC1a	NTC1a	NTC1a
13	NTC1b	NTC1b	NTC1b

Table 5 - Feedback Wiring

All cables and components used for connecting the equipment must comply with prevailing Standards.

When assembling the equipment, be sure it cannot be damaged in any way by the moving components.



Warning

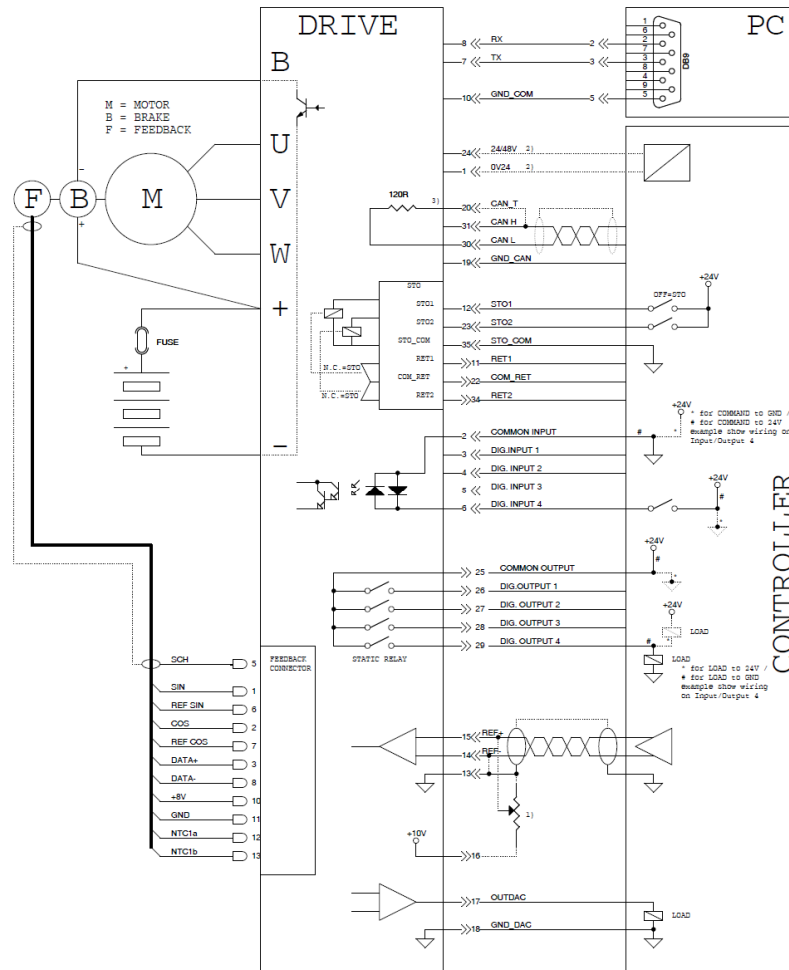
This information is needed to avoid a safety hazard, which might cause bodily injury or death as a result of incorrect operation:

Avoid Ground loops when connecting I/O devices

- When connecting a PC, encoder, switch or actuators on the I/O connector, be very careful that you do not create a path of current that flow from the ground pins on the I/O connector and the -V Main Power Supply terminal. Should the controller's main Ground wires (thick black) or terminals be disconnected while the +VPower Supply wires (thick red) or terminals are connected, the high current would flow from the ground pins, potentially causing serious damage to the controller and/or your external devices.

- Do not connect a wire between the I/O connector ground and the battery minus terminal. Look for hidden connection and eliminate them. Note that signals GND of I/O connector pin N.1,10,13,18,19, are connected to -V Main Power terminal of the drive.
- Have a very firm and secure connection of the controller ground wire and the battery minus terminal.
- Do not use connectors or switches on the power ground cables.
- It is suggested that the external I/O signals are insulated from -Vmain Power Supply to prevent damages.
- Do not allow water to seep into the equipment.

Wiring Example:



- 1) Optional wiring for a potentiometer or a throttle pedal
- 2) Optional wiring for a backup power to logic
- 3) For CAN termination connect pin 20 to CAN H signal (pin31)

1.6 Environmental Condition

You can guarantee the safe operation of the Lafert Smartris Drive by ensuring that it is installed in an appropriate environment.

	Features	Details
	Operating ambient temperature according to IEC 60068-2-2	-10 °C to 40 °C (14 °F to 104 °F)
	Operating ambient temperature with motor derating	40 °C to 55 °C (104 °F to 131 °F)
	Air humidity during operation	Relative humidity < 85%, non-condensing

Installation altitude	< 1000 m above sea level without affecting performance
Protection class	IP30/IP54
Degree of pollution	2 (EN 61800-1, EN61800-5-1)

Table 6 - Environmental Conditions

	Caution
	The Lafert Smartris Drive dissipates its heat by convection or by conduction. The maximum ambient operating temperature of 55 °C (131 °F) must not be exceeded.

	Storage	Details
	Storage Environmental Ambient temperature	-10 °C to 70 °C (14 °F to 158 °F)
	Air humidity during operation	Relative humidity < 90%, non-condensing

HOW TO STORE A LAFERT DRIVE

Short-Term Storage & Long-Term Drive Storage Indoor	- Storage at a temperature between -10° / +60° - No condensation allowed (humidity < 90%) - Cover the equipment during installation for prevent infiltration of dust, metallic shavings, water, oil... - No liability is assumed for errors caused by tampering the system or bad storage.
Short-Term Storage (weeks or months)	Protect the drive from the elements, vibration and extreme heat
Long-Term Storage (years)	Location have to be clean, dry, environment with minimal to no vibration, such as away from railroad tracks and major freeways
Out of storage and Into Service	- Remove Dust and Dirt - For a long storage (6 or more months), provide DC bus capacitor regeneration (supply Vbus "+" and "-" at 48Vdc 1h) before starting work.

Table 7 - Storage

1.7 Approvals

1.7.1 CE Conformity

The Lafert Smartris Drive was tested in authorized testing laboratories in accordance with the requirements of this documentation.

The Lafert Smartris Drive is in conformity with the following **EC Directives**:

- Electromagnetic Compatibility (EMC) (2014/30/EU)
- RoHS Directive (2011/65/EU)
- WEEE Directive (2012/19/UE)

1.7.2 Safety

EN 61800-5-1 Adjustable speed electrical power drive systems - Part 5-1: Safety requirements - Electrical, thermal and energy.

1.7.3 EMC Requirements

In terms of emission and immunity, the Lafert Smartris Drive fulfills the requirement for the category "second environment" (industrial environment).

EN 61800-3 - Adjustable speed electrical power drive systems - Part 3: EMC requirements and specific test methods.

1.7.4 Safety Conformity (STO)

The Lafert Smartris Drive provides a two-channel, functionally safe STO function (Safe Torque Off). The function disables the PWM and the drive can be switched safely to torque OFF.

The circuit design has been tested and subsequently assessed by TÜV Süd. According to that assessment, the circuit design used for the "Safe Torque Off" safety function in the Lafert Smartris Drive is suitable for meeting the requirements for in accordance with

- **EN61508** - Functional safety of electrical/electronic/programmable electronic safety-related systems
- **EN61800-5-2** and category ... – Adjustable speed electrical power drive systems – Part 5-2: Safety requirements – Functional
- **EN ISO 13849-1:2015** - Safety of machinery — Safety-related parts of control systems — Part 1: General principles for design.

The subsystems (Lafert Smartris Drive) are fully described in terms of safety by the following characteristics:

EN 13849-1	EN 61508	PFHD [1/h]
PLe	SIL3	

1.7.5 Environmental Conditions for Vibration/Shock

The drive fulfills the following specifications:

1. Vibration according to DIN EN 60068-2-6:2008
 - **Sweep1 (EN 60068-2-6)**
Frequency range $20\text{Hz} < f < 500\text{Hz}$ 50m/s^2
Acceleration 1Oct/min
50 cycles
 - **Sweep2 (2006/42/EC)**
Frequency range $10\text{Hz} < f < 57\text{Hz}$ 0.075mm
Acceleration 1Oct/min
1 cycle
2. EN60721-3-5: Classification of Environmental Conditions Part 3: Classification of Groups of Environmental Parameters and Their Severities
 - $10\text{Hz} < f < 30\text{Hz}$ $+6\text{dB/Oct}$
 - $30\text{Hz} < f < 200\text{Hz}$ $0.005\text{g}^2/\text{Hz}$
 - $200\text{Hz} < f < 500\text{Hz}$ -12 dB/oct
 - 30 minutes

1.8 Motor/Circuit Protection

Circuit breaker hardware to protect the motor is not required as the motor is protected from overloading by an I²t function in the software.

The parameters for this function are:

- Nominal Current

- Peak Current
- Time of overload

Nominal and Peak Current are characteristics of motors, Time is written in a default value that depends to load and motor but is programmable by user.

It is also programmable the behavior of Drive after Overload time:

- 1) Drive goes in Overload Alarm
- 2) Drive gives at maximum Nominal Current

There is also a function that protect motor to over temperature by a motor temperature sensor: also in this case is programmable the behavior of drive:

- 1) Drive goes in Motor Overtemp Alarm above sensor threshold
- 2) Reading the sensor is possible to reduce the load and speed to try to reduce motor temperature

1.9 Startup

Startup is prohibited within the scope of the EC directives until it has been determined that the machine/system in which this Lafert Smartris Drive is installed corresponds to the regulations within these directives.

For installation in machines and systems, start of intended use of the Lafert Smartris Drive is prohibited until it has been determined that the machine or system complies with the provisions of the EC Machinery Directive 2006/42/EC and the EC EMC Directive 2014/30/EU.

For use in residential areas, additional EMC measures are necessary.

It is the responsibility of the manufacturer of the machine or system to ensure that the threshold values, as stipulated by the EMC regulations, are adhered to.

1.9.1 Correct Use

The Lafert Smartris Drive are intended for operation of permanent magnet synchronous servo motors with compatible feedback systems in stationary machines and systems.

Installation of the Lafert Smartris Drive is only approved in industrial environments. For use in residential areas, additional EMC measures are necessary. The user must prepare a hazard analysis of the final product.

	Caution
	Other uses must first be approved by the manufacturer.

1.9.2 Improper Use

The Lafert Smartris Drive are not suitable for operation of motors other than synchronous servo motors or motors with non-compatible feedback systems.

In addition, the following applications are expected from intended use.

The installation of drives in areas at risk, where inflammable substances or combustible vapors or powders are present, could trigger fire outbreaks or explosions. As such, install the drives far away from said areas at risk, even if they are used with motors fit for use under these conditions.

1.10 Installation Space

A Smartris Drive in IP30 design should be installed in a space that will ensure reliable operation of the drive.

The installation space must meet the requirements of protection class IP54 at least and needs a space of 10mm to other devices.

1.11 Maintenance

Maintenance

The Smartris Drive are maintenance free. Opening the Smartris Drive voids the warranty.

Repairs

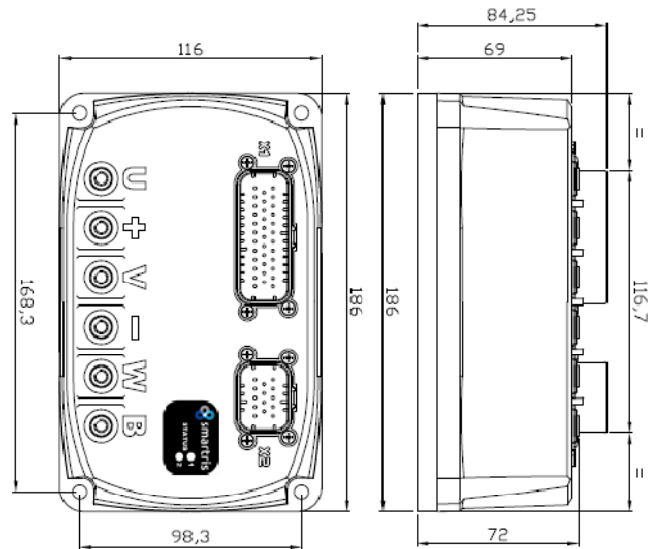
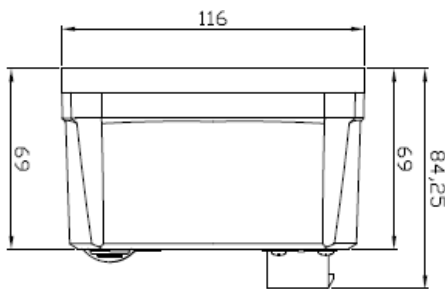
The Smartris Drive may only be repaired by the manufacturer. Opening the Smartris Drive voids the warranty and safety according to the specified standards is no longer ensured.

2 | DIMENSIONS

2.1 Size B Dimension

MECHANICAL SPECIFICATIONS	
Weight [kg]	1,6
Height without plug [mm]	69
Height with plug [mm]	84,25
Width [mm]	186
Depth [mm]	116

Table 8 - Dimensions



Picture 2 - Dimensions

All measures are in mm.

2.2 Mechanical Installation

Safety Instructions:

- For the mechanical installation, the ESD instructions must be observed.
- The drive must be protected from fog, water and penetration of metallic dust in the electrical cabinet.
- The drive must be checked for mechanical damage before installation.
- During installation in an electrical cabinet, sufficient ventilation must be provided.
- The operation of drive exposed to condensation is not permitted.

2.3 Electrical Installation

Safety instructions. For the electrical installation, the ESD instructions must be observed:

	Caution
	• Systems that are electrically connected must be properly secured so they cannot be switched back on and warning signs must be put up. • Before startup, it must be checked that the wiring is correct and is free of mechanical damage. Only drive with wiring in flawless condition may be put into operation. • Incorrect voltage, reverse polarity and nonconforming wiring can damage or destroy the drive. • Excessive or inadequate protection of the power supply can damage the cables or the drive. • Observe the separate instructions for the STO safety function.

2.4 Requirements for Power adapters and Supply Voltage

The STO’s 24 VDC logic supply and 48 VDC power supply must be provided voltage sources with safe extra low voltage in or supply voltage sources having only

	Caution
	The use of unsuitable power adapters not in SELV/PELV design can lead to dangerously high voltages in the even resulting in injuries or death.

The 48 Vdc power adapter supply must permit backfeed up to 60 Vdc, as the drive can generate a voltage up to 60 Vdc at the power adapter terminals when operating as a generator. The power adapter should be designed for such operation.

Otherwise suitable action must be taken to prevent backfeed.

3 | TECHNICAL DATA

3.1 Sizes Main Power Supply

48Vdc (60Vdc Max)

ID	AGV Payload [kg]	MOTOR POWER [W]	MOTOR CABLE [mm ²] - AWG	SUPPLY CABLE [mm ²] - AWG	FUSE [A]
1	600	430	2,5 -AWG14	2,5 -AWG14	16
2					
3					
4	1000	600	6 -AWG10	4 -AWG12	32
5					
6					
7	1500	1000	6 -AWG10	6 -AWG10	63
8					
9					
10	2500	1500	6 -AWG10	6 -AWG10	63
11					
12					

Table 9 - Technical Data

N.B. the list above is an example: the drive is available to use with Lafert or other motor types with similar characteristics.

This manual is available for the following feedback:

- Resolver
- Incremental Encoder
- SinCos Encoder

4 | OPERATION MODE FUNCTIONS



Caution

Smartris Drive has a Safety Torque OFF (STO), so verify that this circuit is correctly supplied (see STO Chapter) before all operation functions.

4.1 Drive Control Mode

The drive can be piloted in 2 different mode:

- Analog Mode
- CANOpen Mode



Information

Note: the Drive Control Mode is set by a internal parameter. To change the Control Mode please contact the manufacturer

4.2 Analog Mode

In this operation mode the Drive can be piloted with an analog reference -10 Volt +10Volt. Giving voltage on +VREF and -VREF (see paragraph Connector X1) it is possible to supply to the drive the speed set point.

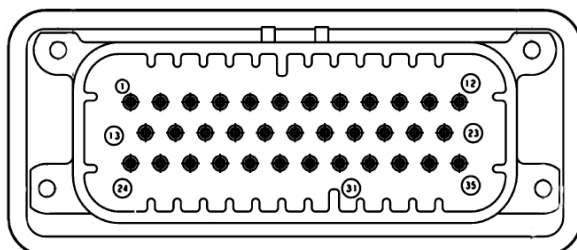
The motor speed will depend from the reference voltage given on +VREF and -VREF and from the maximum speed available on the drive.

The input range of the reference voltage for the Speed Set Point is from -10V to +10V.

The Speed Set Point is proportional to the voltage supplied on the concerned input.

Giving a +10V voltage the motor will set to the maximum speed in clockwise rotation, motor front, while giving a -10V the motor will set to the maximum speed in counter clockwise rotation, motor front.

Example		
Input Voltage	Rotation	Speed
+10V	clockwise	+ 3000 r.p.m
-10V	counter clockwise	- 3000 r.p.m
+5V	clockwise	+ 1500 r.p.m
-5V	counter clockwise	- 1500 r.p.m



X1 - I/O SIGNAL		
PIN	SIGNAL	DESCRIPTION
3	IN1	Digital Input 1 - RUN COMMAND
4	IN2	Digital Input 2 - STOP COMMAND
5	IN3	Digital Input 3 - ELECTRONIC DYNAMIC BRAKE COMMAND (see Par 4.5.2)
6	IN4	Digital Input 4 - ELECTRONIC RESET DRIVER
13	GND_REF	GND Analog Reference
14	REF -	Analog Input Neg
15	REF +	Analog Input Pos
16	+10V	10 Vdc Output

Table 10 - IO Analog Mode

To move the motor is necessary that the inputs IN1 and IN2 are ON.

IN1	IN2	DRIVE STATE	STATUS 1 LED GREEN	STATUS 2 LED YELLOW	LED VIEW
OFF	X	STAND-BY	"BLINK"	OFF	 1 BLINK - 50%  2 OFF
X	X	FAULT – See Diagnostic Chapter	"BLINK" code [x]	"BLINK" code [y]	 1 see code (Chap.7)  2
ON	ON	RUN	ON	OFF	 1 ON  2 OFF
ON	OFF	STOP	ON	ON	 1 ON  2 ON
X	X	SAFETY – STO OFF (see STO Chapter)	OFF	"BLINK"	 1 OFF  2 BLINK - 50%

Table 11 - Analog Mode Led Status

IN3 is programmable input for the logic *Electronic Dynamic (Emergency) Brake*; it is possible to enable/disable this function and select the logic:

- 1) IN3 ON→OFF activate the Electronic Dynamic Emergency Brake
- 2) IN3 OFF→ON activate the Electronic Dynamic Emergency Brake

To move the motor is necessary that the inputs IN1 and IN2 are ON. See Digital I/O chapter

Variable Monitoring

In Analog Mode it's possible to monitor a list of variables on drive by CANOpen, connecting a CAN Interface and using LafertDrive SW:

- Object 2002h: Drive Status Mode
- Object 2003h: Warning
- Object 2004h: State Lafert Servo Drive Machine
- Object 2030h: Temperature Drive
- Object 2031h: Temperature Motor
- Object 2032h: Temperature Heat Sink
- Object 2041h: Voltage Bus
- Object 2050h: Torque Current
- Object 2051h: Power Drive
- Object 2052h: Power Motor
- Object 2053h: Velocity Filtered

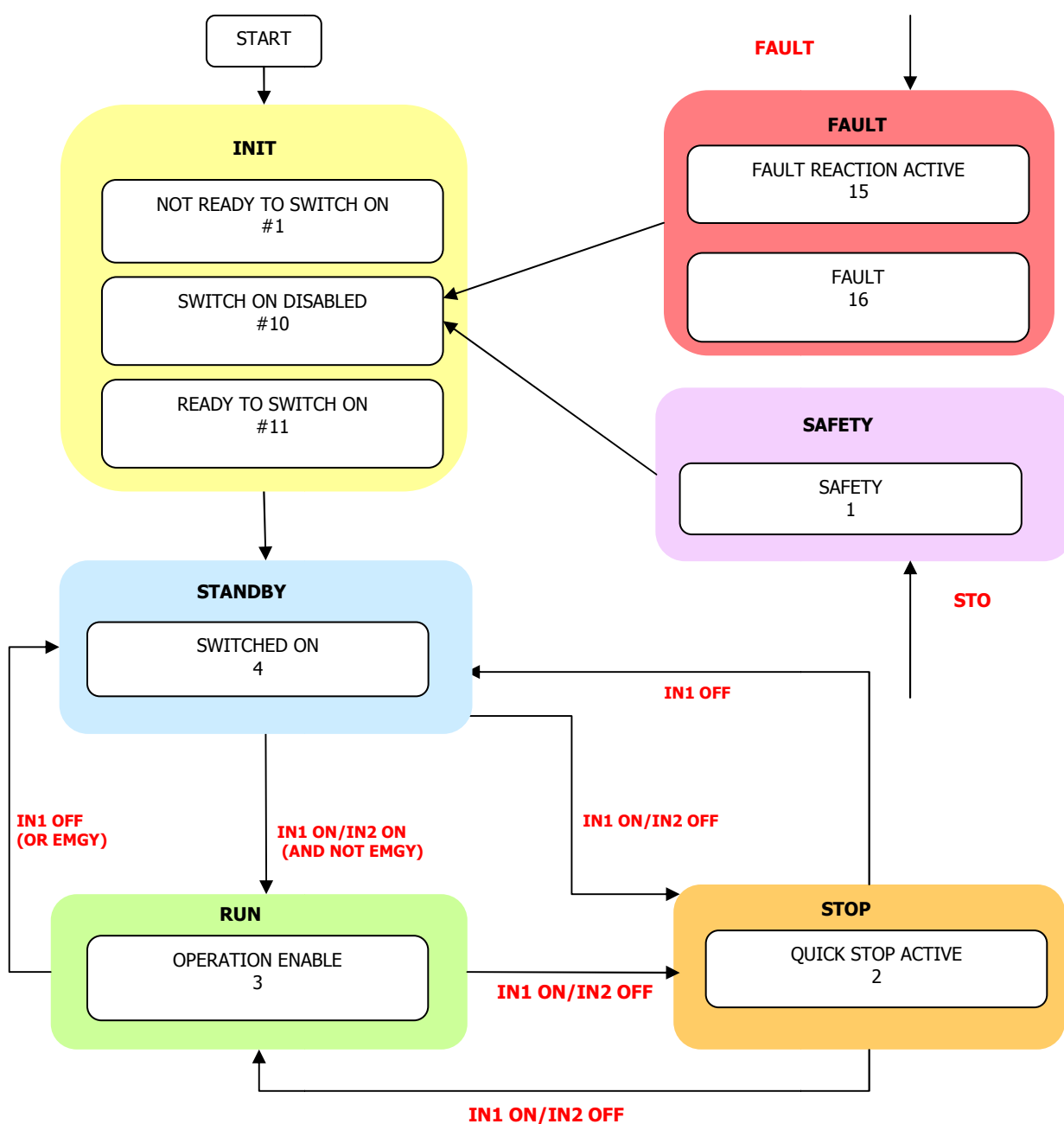
Smartris User Guide - FW 1.0.8

- Object 3020h: Drive Digital Input
- Object 3022h: Analog Input
- Object 4000h: Safety State

DAC monitoring

It is possible to configure the analogue output as a monitoring. The object to set the DAC configuration is the 3050_h

- Analog Output = 0: Disabled
- Analog Output =1: Set as General Purpose
- It is possible to set a Digital value in the sub-index 2 and reading in the output.
- Read analog Output (object 3050_h: 3)
- Example: Analog Output configured as "Velocity Monitoring" value (object 3050_h: 2)



Picture 3 - Analog Mode State Machine

4.3 CANOpen Mode

The Smartris Drive is available with CANOpen (DS301-DS402) alternatively to the Analog Mode: see the CANOpen Chapter for instructions.

To use CANOpen Smartris connected to laptop (free PC Tools available on the WEB) a USB to CAN interface is needed and one suggested is:

PCAN-USB - IPEH-002021 (Peak manufacturer)

<https://www.peak-system.com/PCAN-USB.199.0.html?&L=1>



See also "Lafert CANOpen Manual"

4.4 Ramp Speed Set-up

It's possible to set the drive in ramp mode. This Operation mode makes that the variation of speed can be defined by ramp defined by user.

Enable of ramp on speed set point is performed by a flag on *Drive Parameters (contact Manufacturer)*

Ramp are enabled when this bit is checked. If you enable this bit you must save on E2Prom and then reset drive before to go in run mode.

There are two parameters that rule action of ramp, time to accelerate from 0 to max speed and time to decelerate from max speed to 0.

Ramp up delay is specified by *RampAcc* on *Drive Parameters (contact Manufacturer)*, and ramp down is specified by *RampDec* on *Drive Parameters (contact Manufacturer)*,

If the speed ramps are active, each variation of speed set point will corresponds a ramp with this parameter

These parameters are specified in millisecond, their maximum values are 32000. Minimum value depends by max velocity of drive. If you specify a too low value you will get a parameter program error.

This ramp operation mode not is active with STOP command.

During the RUN state If Run command drive down, the drive STOP motor rotation with rotor torque with ramp programmed in *RampDec*.

4.5 Stop with Ramp

It's possible to set the drive with stop with ramp mode. This Operation mode makes that the variation of speed can be defined by ramp defined by user.

Enable of ramp on speed set point is performed by *Delay1* (*Drive Parameters (contact Manufacturer)*)

Ramp are enabled when this bit is checked. If you enable this bit you must save on E2Prom (with button) and then reset drive before to go in run mode.

There is a parameter that rules action of stop with ramp, time to decelerate from max speed to 0:

ramp down is specified by *Delay2* (*Drive Parameters (contact Manufacturer)*)

If the stop ramp is active, each variation of speed set point will corresponds a ramp with this parameter)

This parameter is specified in millisecond, their maximum values are 32000. Minimum value depends by max velocity of drive. If you specify a too low value you will get a parameter program error.

This Stop with Ramp operation mode is independent of Speed with Ramp mode operation.

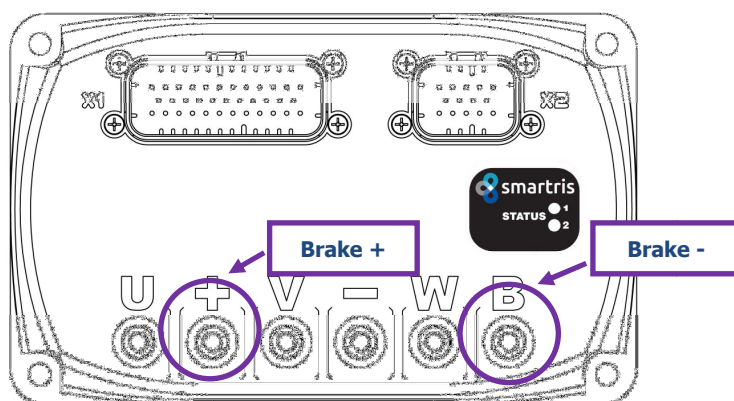
4.6 Motor Brake

The Motor Brake Management is an output who can drive and supply power directly to a motor brake.

	It is possible to use DOUT4 - Digital Output 4 – to drive a relay for brake management with externally power supplied
---	---

To enable this function, you must check BIT_BRAKE (second from the top) into window mask Customer Flag. The brake can be set in Automatic or Manual Mode:

- **Automatic Mode:** The brake will be released (Brake+ = 24V) automatically when the Smartris Drive is set in RUN/Enable.
- **Manual Mode:** the user can be release the brake using a dedicated command in CANOpen/Parameter in RAM.

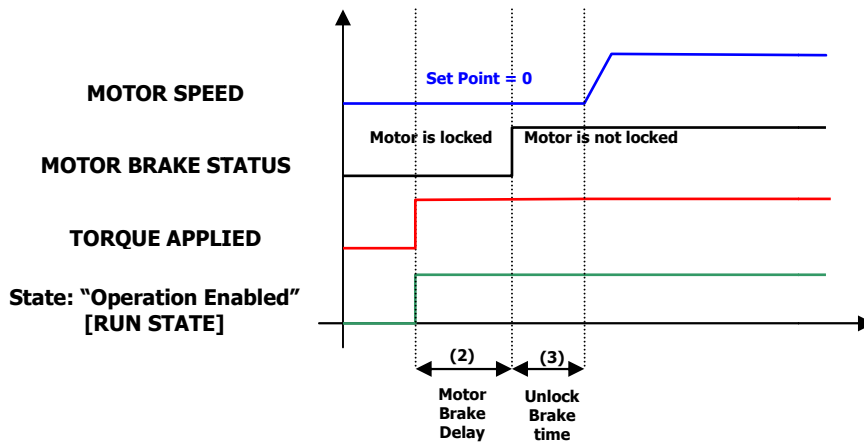


Brake Options			
N°	Field	Configuration	Definition
1	Motor Brake Option (*)	0 _b 1 _b	Motor Brake disabled or Motor Brake is not present Motor Brake enabled
2	Motor Brake Delay	[ms * 10]	Delay open command. This timeout is the delay between STBY Status and unlock brake.
3	Unlock Brake time	[ms * 10]	Delay between STOP and RUN mode before unlock Brake. This timeout depends by kind of motor brake.
4	Brake timeout	[ms * 10]	Only without Dynamic Brake (see object 0x3007) Max time programmed for natural Inertia deceleration. At the end of this timeout the brake is locked and drive will be in STBY status.
5	Automatic/Manual Mode Configuration	0 _b 1 _b	Automatic Mode Activated Manual Mode Activated
6	Motor Brake Status	0 _b 1 _b	Brake Status: activated → Motor is locked Brake Status: released → Motor is not locked
7	Brake Type	1 2	Magnetic Brake Spring Brake

Table 12 – Brake Options

	(*) If the motor does not have the brake, this value has 0 as default value. <u>The user CAN'T enable because it is not present.</u>
---	--

AUTOMATIC MODE



Picture 4 - Brake Automatic

MANUAL MODE

In Analog mode set a flag on *Drive Parameters (contact Manufacturer)* "MANUAL BRAKE" To command the brake use **ParComBrake**

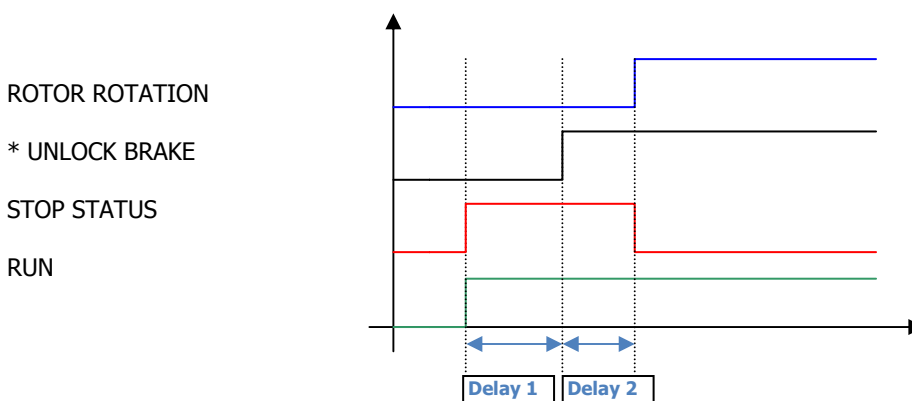
- **ParComBrake** bit3 = 1 BRAKE released → motor axis free
- **ParComBrake** bit3 = 0 BRAKE active → motor axis locked

In CANopen Mode set **bit 1** on 60FEh subindex 1,

- **SetBrake** [index 60FEh, subindex 1, bit 0] = 1 BRAKE RELEASED

4.6.1 Phase STBY → RUN

In STBY status the brake is turn-off and rotor lock. Enabling RUN the drive will stay in STOP status for the programmed time in Delay1, at the end of this time the rotor brake will be turn-on, but the drive will be in STOP status for an additional delay programmed in Delay2: it have to be programmed with the real relay and brake delay. At the end of this second delay time the drive will be in RUN mode and it's ready to drive the motor with speed set point supplied. In STOP status the motor is in standstill torque.



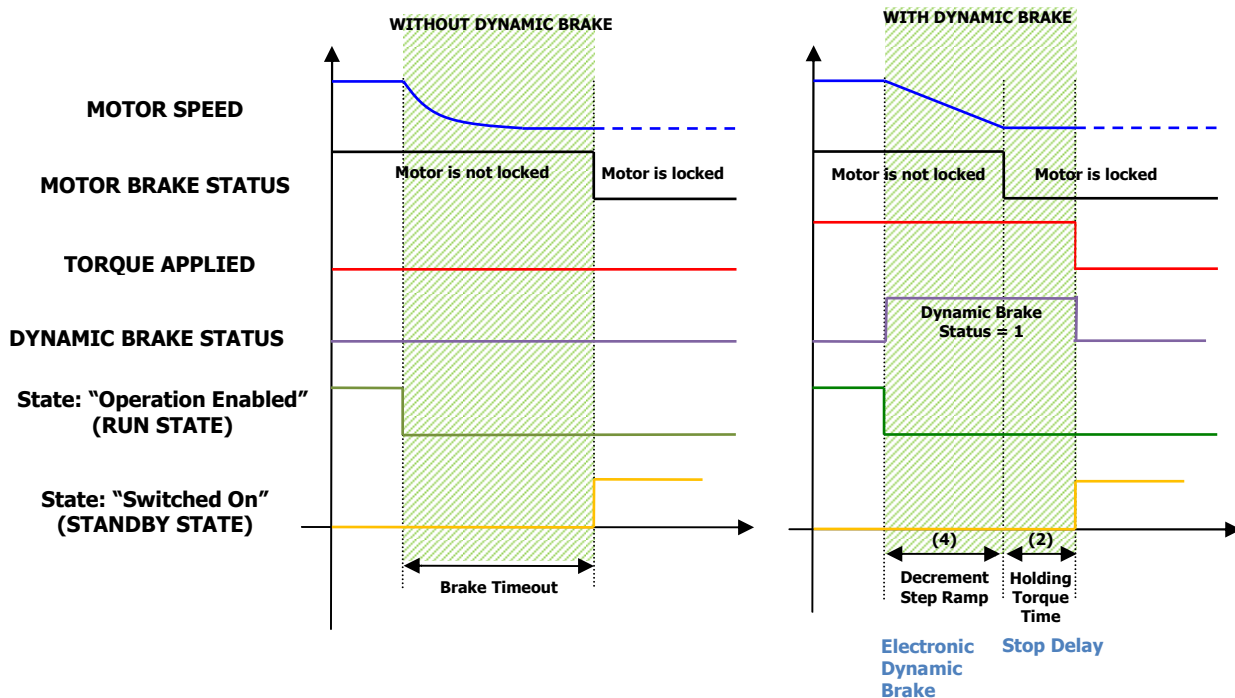
Picture 5 - Delay time Brake Automatic

4.6.2 Phase RUN->STBY

In RUN status the brake is unlock.

In the first case when *electronic dynamic brake function* is ON, the motor will decrement speed with a ramp set by *electronic dynamic brake* parameters. When the ramp finish, the drive lock brake motor and stay in STOP for a delay time programmed with *DynBrake* parameter, finally the drive will turn in STBY mode with locked brake.

If *electronic dynamic brake* is off, the drive will decrease speed with natural inertia. When the speed is zero the drive lock motor brake, but if deceleration speed is major than delay time in set a flag on *Drive Parameters (contact Manufacturer)*, the drive however lock brake.



Picture 6 - Dynamic Brake

Disabling RUN the drive action will be different depending by activation of Electronic Dynamic Brake function.

4.7 Digital I/O

In Lafert Drives are available:

- 4 Digital Inputs **DIG-INx**
- 4 Digital Outout **DIG-OUTx**
- 2 Safety Digital Input **DIG-STO**

4.7.1 Digital Input

When is unset the flag "Enable CAN" (flag on *Drive Parameters (contact Manufacturer)*) the drive is in "ANALOG MODE" so to move the motor are used 2 digital inputs:

- **DIG-IN1** = RUN in Analog Mode

- **DIG-IN2** = STOP in Analog Mode

If it is available the STO Safety: if the STO is active the drive goes in SAFETY status independently to other selection.

In case of FAULT the drive goes in FAULT State.

Instead if is set flag "Enable CAN" the State Machine follows the "controlword" (6060h) and Digital Input are ignored.

DIG-IN3 can be programmed as "**Enable Input Emergency**".

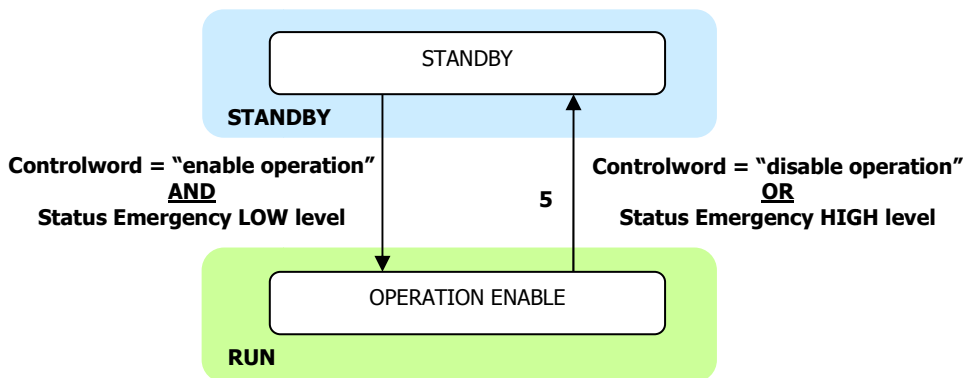
In this case DIG-IN3 is used to go from RUN to STANDBY in emergency condition with dynamic brake.

It is possible to enable this function:

1. SERIAL: setting flag on *Drive Parameters (contact Manufacturer)*
2. CANOPEN: setting object 0x3008h (Emergency Input Enable) subindex 1

There is the possibility to select the negative front to select the Dynamic Brake:

1. SERIAL: setting flag on *Drive Parameters (contact Manufacturer)*
2. CANOPEN: setting object 0x3008h (Emergency Input Enable) subindex 2



Picture 7 - Emergency enable configuration

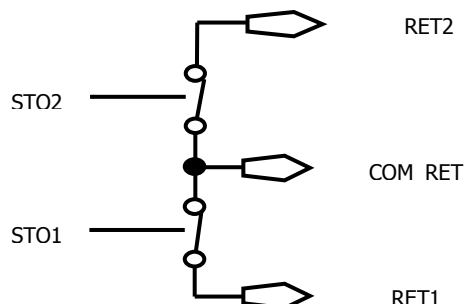
The digital input number 3 can be configured as enabling signal hardware to move from "Operation Enable" state [RUN] to "Switched On" state [STANDBY].

It can be considered as emergency signal but it isn't safety certificated (for disabling the power in safety certificated condition referred to STO chapter of Drive User Guide).

DIG-IN4 is Reset Drive Hardware.

4.7.2 Digital Input Safety STO

DIG-STO: STO input follows the circuit



Picture 8 - STO Output FeedBack Relay

If an application requires controlled braking before using the STO function, the drive must be braked first and the STO function must be activate with a delay:

1. Controlled braking of drive
2. Once standstill is reached, disable the drive
3. In the case of a suspended load, mechanically lock the drive as well
4. Activate STO function.

	Caution The Drive cannot hold the load with the STO function activated because the motor no longer supplies any torque. • If the STO function is activated during operation, the drive will stop in an uncontrolled manner. • If the drive has the Safety Torque OFF (STO), verify that this circuit is correctly supplied before all operation functions.
---	--

Two input feedback

Every single output relay is read.

Safety controller can read every single command corresponding to every relay output: fully monitoring of safety functions.

It is possible with CANopen DSP402 read the status of Digital Inputs with 0x60FD.

4.7.3 Digital Output

DIG-OUT1: shows if the drive is:

- '1' = Drive OK
- '0' = Drive in FAULT

DIG-OUT2 e DIG-OUT3 are free and programmable

DIG-OUT4 shows the BRAKE status:

- '1' = brake released, motor axis free
- '0' = brake active, motor blocked

It is possible with CANopen DSP402 read the status of Digital Outputs with 0x60FE

5 | STO (SAFE TORQUE OFF)

The Safety Circuit provides drive integrated safety function according to IEC 61800-5-2:2016. STO (Safe Torque Off) Module prevents the creation of torque in the motor and corresponds to an uncontrolled stop in accordance with stop category 0 of IEC 60204-1.

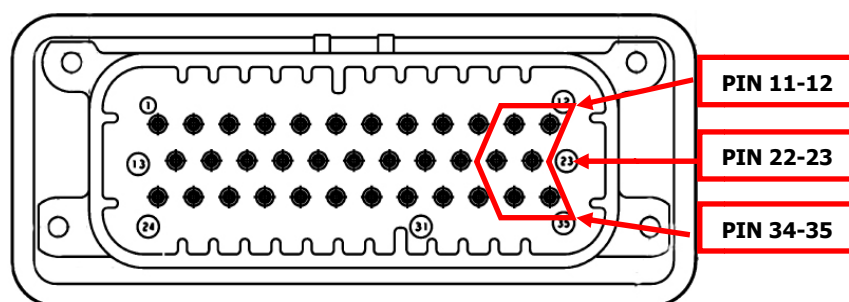
5.1 STO (Safe Torque Off)

Power, that can cause rotation (or motion in the case of a linear motor), is not applied to the motor. In STO condition, the drive will not provide power to the motor which cannot generate torque.

Note:

1. This safety function corresponds to an uncontrolled stop
2. This function may be used where power removal is required to prevent an unexpected start-up.
3. In circumstances where external influences (i.e., falling of suspended loads) are present, additional measures (i.e., mechanical brakes) may be necessary to prevent any hazard.

5.1.1 STO Connection



I/O SIGNAL		
PIN	SIGNAL	DESCRIPTION
11	RET1	Safety Relays 1 (Output)
12	STO1	Safety Relays STO1 (Input)
22	COM_RET	Common Relays
23	STO2	Safety Relays STO2 (Input)
34	RET2	Safety Relays 2 (Output)
35	STO_COM	Common STO1 / STO2 (Input)

Table 13 - STO Wiring

5.2 Hardware Specifics

STO Input	Data
STO Inactive status (Normal operation) Input voltage	20...28 Vdc
STO Active (SAFETY) input voltage	< 2.4 Vdc
Input current (Typ)	29 mA
Active response time (time between normal operation to STO Active)	10 mS
Contact specification RET1, RET2 (rated voltage/current)	30V / 0.5A

Are available 2 separated inputs STO-1 e STO-2 with reference STO-COM, and 2 separated relays output (N.C.) RET-1 and RET-2, connected.

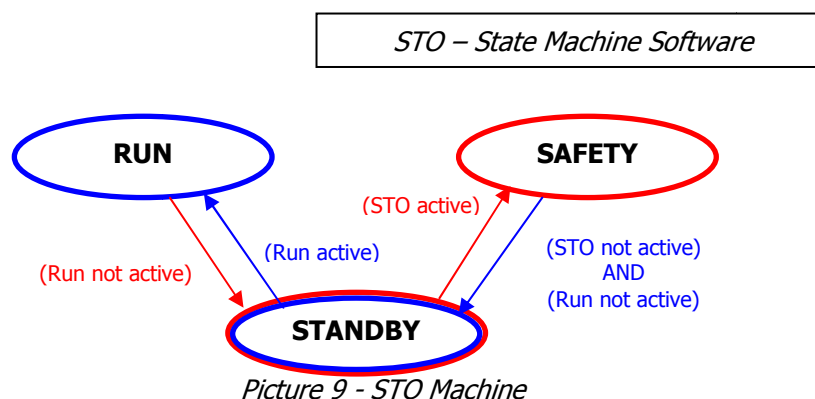
Input 1	Input 2	Output 1	Output 2	Output SW
STO1	STO2	RET1	RET2	STATUS
0	0	CLOSE	CLOSE	SAFETY
24V	0	OPEN	CLOSE	SAFETY
0	24V	CLOSE	OPEN	SAFETY
24V	24V	OPEN	OPEN	NORMAL MODE

Table 14 - STO Output State

SAFETY status (no torque condition) is active when at least one input between STO1 and STO2 is 0V (not supplied).

5.3 Software Specifics

Correct procedure to SAFETY is: RUN → STANDBY → SAFETY.



	Caution To reactive the standby status is mandatory that STO and RUN are not active and do the procedure - in Analog Mode: SAFETY → STANDBY → RUN - in CANopen Mode: SAFETY → "SWITCH ON DISABLED" → "READY TO SWITCH ON" → "SWITCH ON" → "OPERATION ENABLED"
---	--

5.4 Safe Operation Sequence Procedure

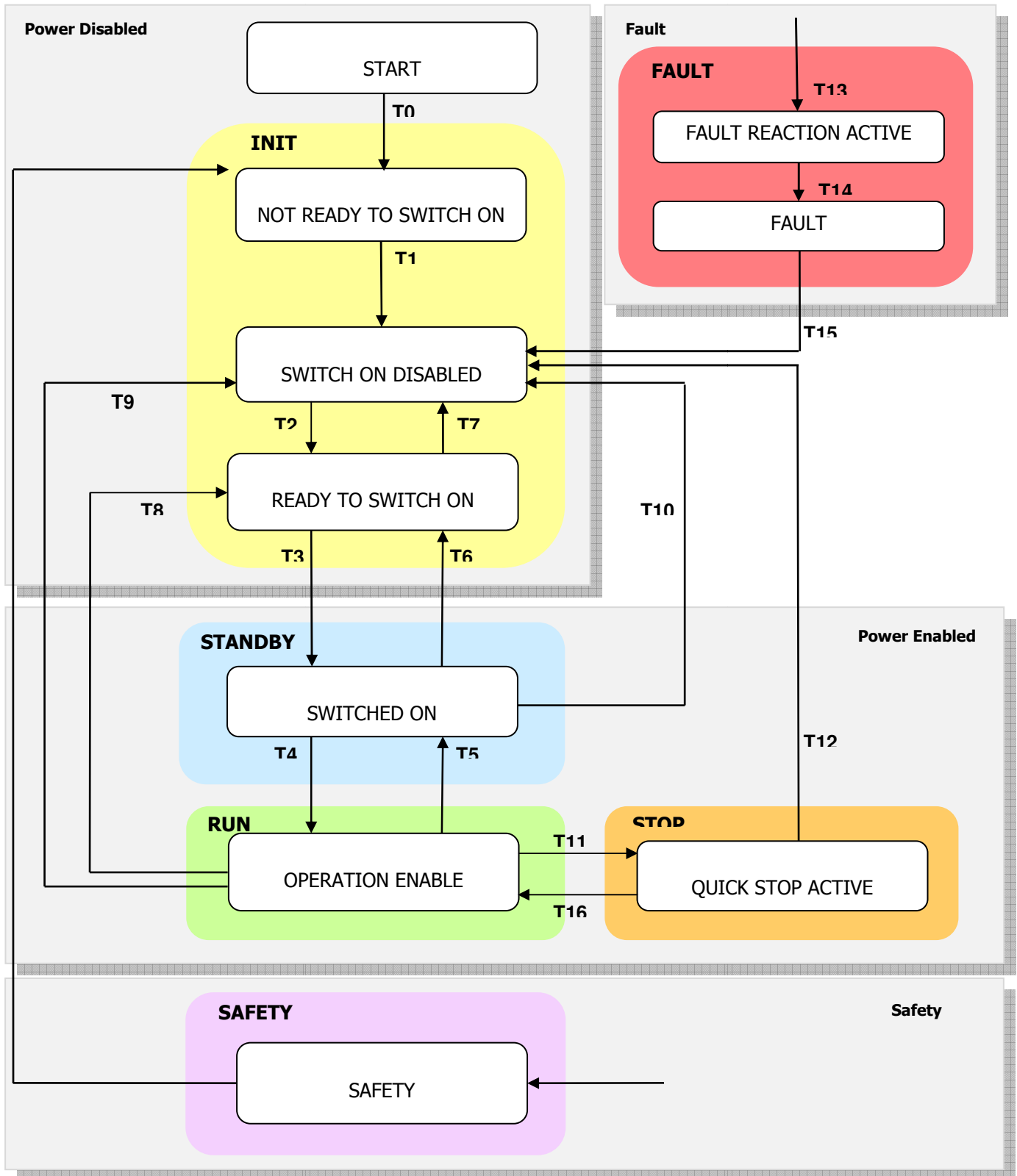
If an application requires controlled braking before using the STO function, the drive must be braked first and the STO function must be activate with a delay:

1. Controlled braking of drive
2. Once standstill is reached, disable the drive
3. In the case of a suspended load, mechanically lock the drive as well
4. Activate STO function.

	Caution The Smartris Drive cannot hold the load with the STO function activated because the motor no longer supplies any torque. If the STO function is activated during operation, the drive will stop in an uncontrolled manner.
---	--

5.5 State Machine DSP402 with Safety State

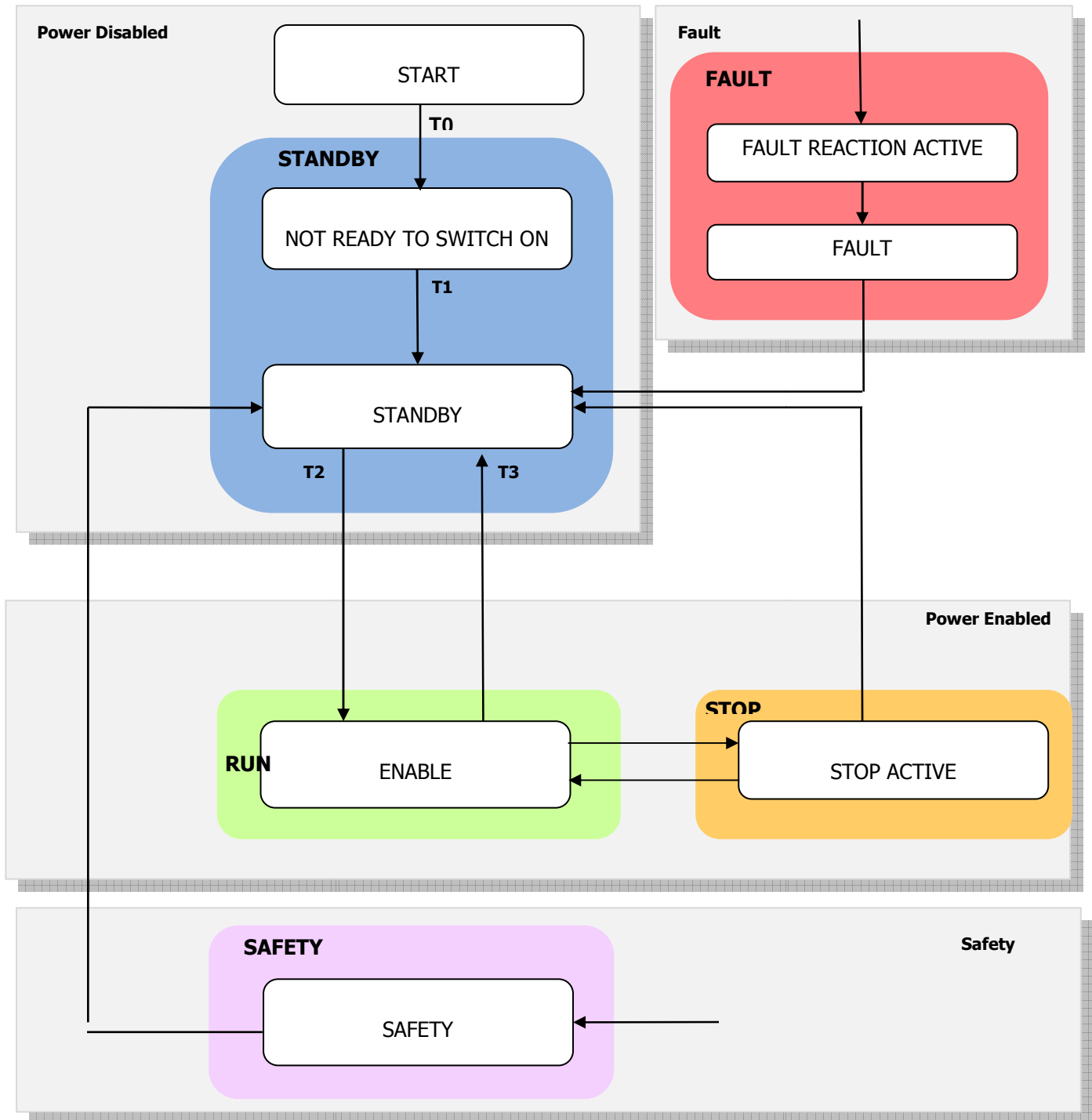
The following picture shows the safety state. This state is added in the state machine DSP402. To exit by Safety State it is necessary to send the controlword with value "Disable Voltage".



Picture 10 - State Machine DSP402 with Safety State

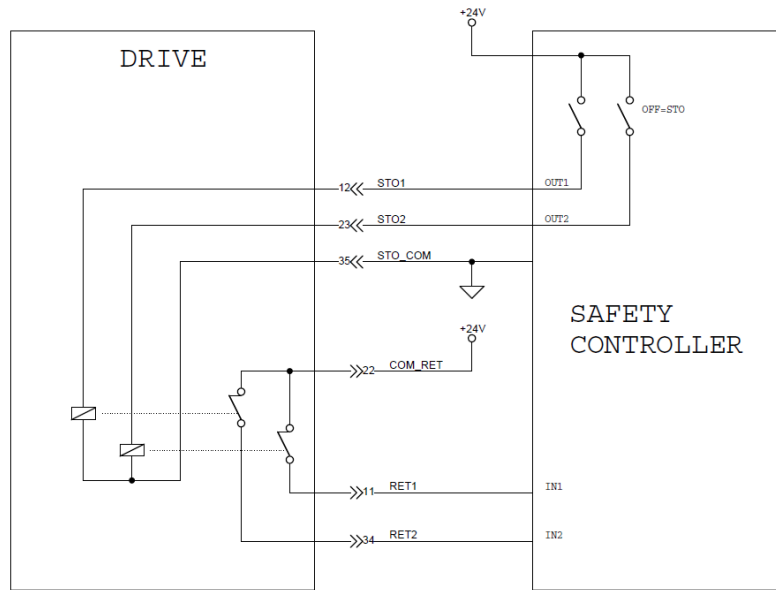
5.6 State Machine Analogue Mode with Safety State

The following picture shows the safety state in Analogue Mode



Picture 11 - State Machine Analogue Mode with Safety State

5.7 Example of Schematic Application



Picture 12 - STO Application

Two input feedback

Every single output relay is read.

Safety controller can read every single command corresponding to every relay output: fully monitoring of safety functions.

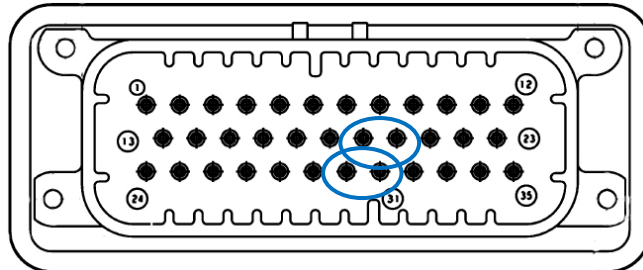
5.8 Functional Check

The STO function must be checked at initial startup, after any work on the system's wiring, and after replacing one or more components of the system.

The circuit needs also a verify every 6 months of correct functioning

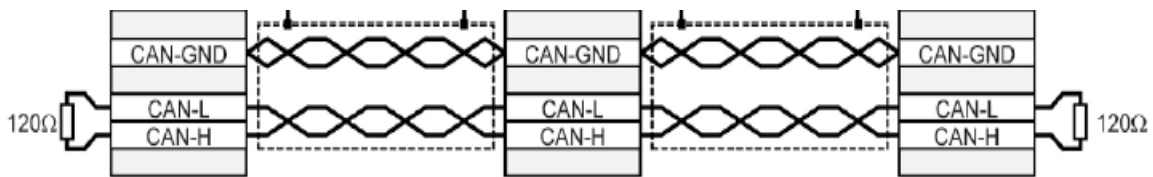
6 | CANOPEN COMMUNICATION

6.1 CANopen Network Topology Overview



CANOpen SIGNAL		
PIN	SIGNAL	DESCRIPTION
19	GND_CAN	GND reference for CAN
20	CAN_T	120 Ω Termination resistance CAN (connect to CAN H)
30	CAN_L	CAN_L Connection
31	CAN_H	CAN_H Connection

Table 15 - CANOpen Signal Wiring



Picture 13 - CANOpen Connection

All nodes of a network are principally connected in series, so that the CAN cable is looped through all controllers.

The two ends of the CAN cables have to be terminated by a resistor of 120 Ω +/-5%. The CAN bus can be closed with a resistance terminator into the drive using Pin 20. The CAN_T pin has to connect to CAN_H pin.

For cabling shielded cable with exactly two twisted pairs have to be used.

- One twisted pair is used for CAN-H and CAN-L.
- One twisted pair is used commonly for CAN-GND.

For further information refer to the Controller Area Network protocol specification, Ver. 2.0, Robert Bosch GmbH, 1991.

Technical data CAN bus cable:

- 2 twisted pairs, $d \geq 0,22 \text{ mm}^2$ shielded
- loop resistance $< 0,2 \text{ } \Omega/\text{m}$
- char. impedance 100-120 Ω

Client - Server relations

A CAN master (or client) is a controller that makes requests to nodes to respond to its commands. A CAN slave (or server) responds to the commands issued by the CAN master. The CAN protocol permits both single-master and multiple-master networks.



Information

The Lafert Servo Drive is the SLAVE and the machine controller or PLC is the MASTER.

Every servo drive has a unique ID in the range [1...127]. The network master does not require an ID. As a slave, the servo drive never sends an unrequested message, other than emergencies. The drive responds only to messages addressed to its ID or to broadcast messages, which have an ID of 0. All messages sent by a servo drive are marked with its own ID.



Caution

If two servo drives have been assigned the same ID, the CAN network may crash.

CANopen Baudrate & Node ID

- Baudrate set by a CANOpen (Default 1000K)
- Node ID set by software (Default 1)

6.2 General Features of CAN

Compliance with the directives CiA DS301 v4.02 and DSP402 v2.0:

- Communication Network set by CANOpen object
- NMT- Network Management Protocol
- SDO - Service Data Object Protocol: upload, download and abort Protocol
- PDO - Process Data Object (4 TPDO, 4 RPDO), event timer, granularity 8 bit
- Emergency Protocol
- Heartbeat and Node guarding (Monitoring Protocol)
- Store non volatile memory (communication + manufacturer + device profile)
- Restore default CanOpen from non volatile memory
- Input/Output management by CANopen Object
- State Machine according by DSP402
- Profile Mode p402 supported: Profile Velocity Mode (see object 0x6502)

6.3 Communication CANopen Object (COB)

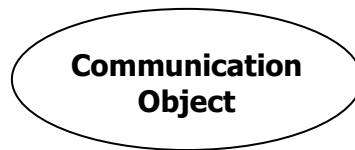
The communication objects are standardized with the DS301 CANopen communication profile. The objects can be classified into 4 groups according to their tasks.

Special Objects

Emergency (**EMCY**)
SYNC (Not implemented)

PDO

TPDO-1 RPDO-1
TPDO-2 RPDO-2
TPDO-3 RPDO-3
TPDO-4 RPDO-4



Network Management

NMT (Services)
NMT (Node Guarding)
NMT (Heartbeat)

SDO

Tx_SDO
Rx_SDO

- PDOs (process data objects) for real-time transmission of process data
- SDOs (service data object) for read and write access to the object Dictionary
- Objects for controlling CAN messages:
 - SYNC object (synchronization object) for synchronization of network devices
 - EMCY object (emergency object), for signaling errors of a device or its peripherals.
- Network management services:
 - NMT services for initialization and network control (NMT: network management)
 - NMT Node Guarding for monitoring the network devices
 - NMT Heartbeat for monitoring the network devices

Object Dictionary

The most important part of a device profile is the Object Dictionary description. The Object Dictionary is essentially a grouping of objects accessible via the network in an ordered pre-defined fashion. Each object within the dictionary is addressed using a 16-bit index.

The general structure of the object dictionary is as follows:

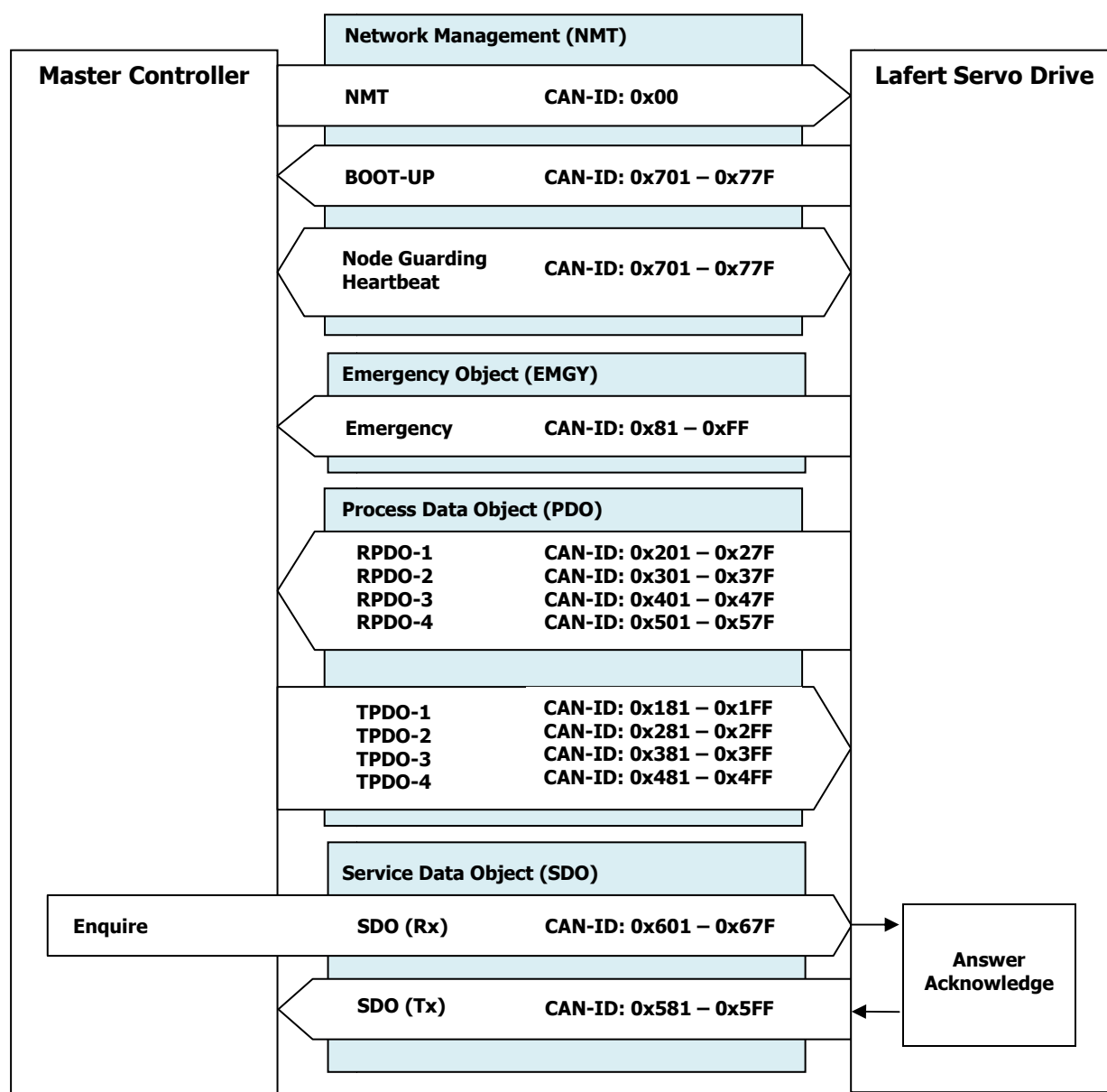
Index, Sub-Index (HEX)	OBJECT (Symbolic Name)	Name	Type	Attribute	M/O
---------------------------	---------------------------	------	------	-----------	-----

- **Index, Sub-Index:** The Index column denotes the objects position within the Object Dictionary. This acts as a kind of address to reference the desired data field. The sub-index is not specified here. The sub-index is used to reference data fields within a complex object such as an array or record.
- **Object:** The Object column contains the Object Name and is used to denote what kind of object is at that particular index within the Object Dictionary.
- **Name:** The name column provides a simple textual description of the function of that particular object.
- **Type:** The type column gives information as to the type of the object. Eg: Boolean, Floating number, Unsigned Integer, Signed Integer etc.
- **Attribute:** The Attribute column defines the access rights for a particular object. Eg: rw (read and write access), wo (write only), ro (read only), Const (read only and value is constant).
- **M/O:** The M/O column defines whether the object is **M**andatory or **O**ptional

The Standard object dictionary is as shown below:

Index (HEX)	Object
0000	Not used
0001-001F	Static data types
0020-003F	Complex data types
0040-005F	Manufacturer specific Complex data types
0060-007F	Device Profile Specific Static Data Types
0080-009F	Device Profile Specific Complex Data Types
00A0-0FFF	Reserved for further us
1000-1FFF	Communication Profile Area
2000-5FFF	Manufacturer Specific Profile Area
6000-9FFF	Standardized Device Profile Area
A000-FFFF	Reserved for further use

For communication between controller and Lafert Servo Drive the following communication objects (COB) are available.



6.4 Network Management (NMT)

The Network Management (NMT) is node oriented and follows a master-slave structure. NMT objects are used for executing NMT services. Through NMT services, nodes are initialized, started, monitored, resetted or stopped. All nodes are regarded as NMT slaves. An NMT Slave is uniquely identified in the network by its Node-ID, a value in the range of [1 to 127].

NMT requires that one device in the network fulfils the function of the NMT Master.

NMT Services:

- **Module Control Services:** Through Module Control Services, the NMT master controls the state of the NMT slaves. The state attribute is one of the values (STOPPED, PRE-OPERATIONAL, OPERATIONAL and INITIALISING). The Module Control Services can be performed with a certain node or with all nodes simultaneously.
- **Error Control Service:** Through Error control services the NMT detects failures in a CAN-based Network. Local errors in a node may e.g. lead to a reset or change of state. Error Control services are achieved principally through periodically transmitting of messages by a device. There exist two possibilities to perform Error Control i.e. Node Guard and Heart Beat Error Control.
- **Bootup Service:** Through this service, the NMT slave indicates that a local state transition occurred from the state INITIALISING to the state PRE-OPERATIONAL.

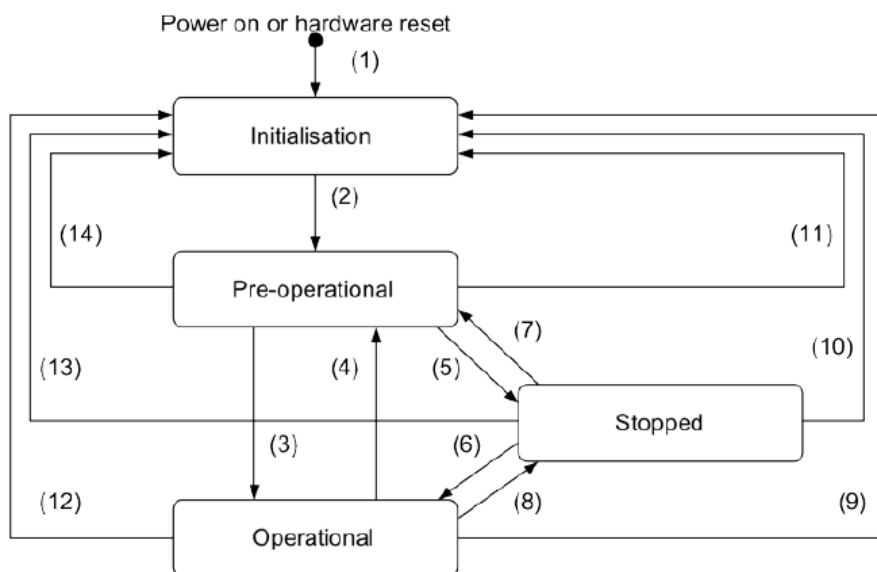
NMT state machine

The following picture shows the NMT state diagram of a CANopen device is specified.

CANopen devices enter the NMT state Pre-operational directly after finishing the CANopen devices initialization. During this NMT state CANopen device parameterization and CAN-ID-allocation via SDO possible. Then the CANopen devices may be switched directly into the NMT state Operational.

The NMT state machine determines the behaviour of the communication function unit.

The coupling of the application state machine to the NMT state machine is CANopen device dependent and falls into the scope of device profiles and application profiles.



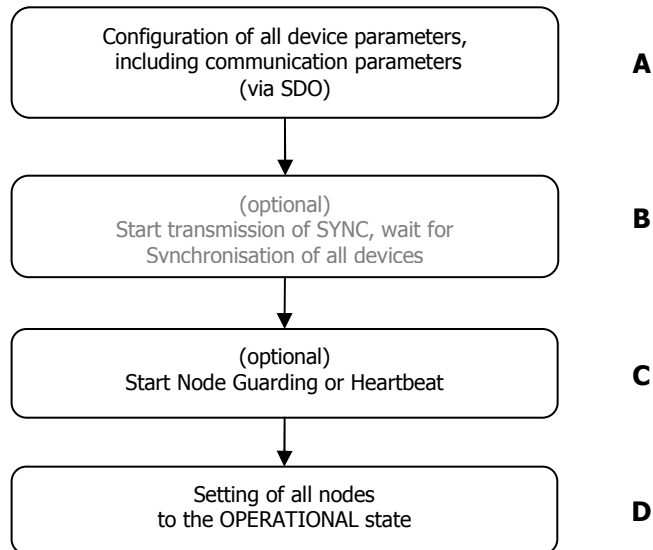
(1)	At Power on the NMT state initialisation is entered autonomously
(2)	NMT state Initialisation finished - enter NMT state Pre-operational automatically
(3)	NMT service start remote node indication or by local control
(4),(7)	NMT service enter pre-operational indication
(5),(8)	NMT service stop remote node indication
(6)	NMT service start remote node indication
(9),(10),(11)	NMT service reset node indication
(12),(13),(14)	NMT service reset communication indication

States and Communication Object Relations:

Table below shows the relation between communication states and communication objects. Services on the listed communication objects may only be executed if the devices involved in the communication are in the appropriate communication states.

	Pre-operational	Operational	Stopped
PDO		X	
SDO	X	X	
SYNC	X	X	
TIME	X	X	
EMCY	X	X	
Node control and error control	X	X	X

Network Initialisation Process:



- STEP A): the device is in the node state PRE-OPERATIONAL which is entered automatically after power-on. In this state the devices are accessible via their default-SDO, the configuration of SDOs settings and optionally the setting of COB-IDs may be performed via SDO objects. In many cases a configuration is not even necessary as default values are defined for all application and communication parameters.
- STEP B): If the application requires the synchronisation of all or some nodes in the network, the appropriate mechanisms can be initiated in the optional Step B (now the SYNC feature is not implemented). It can be used to ensure that all nodes are synchronised by the SYNC object before entering the node state OPERATIONAL in step D. The first transmission of SYNC object starts within 1 sync cycle after entering the PRE-OPERATIONAL state.
- STEP C): In this step the Node guarding or Heartbeat can be activated using the guarding parameters configured in step A.
- STEP D): Now the master controller has to move the drive in OPERATIONAL state. With step D all nodes are enabled to communicate via their PDO objects.

6.5 SDO and PDO Protocol

The Lafert Drive can be parameterized via SDOs and controlled via PDOs.

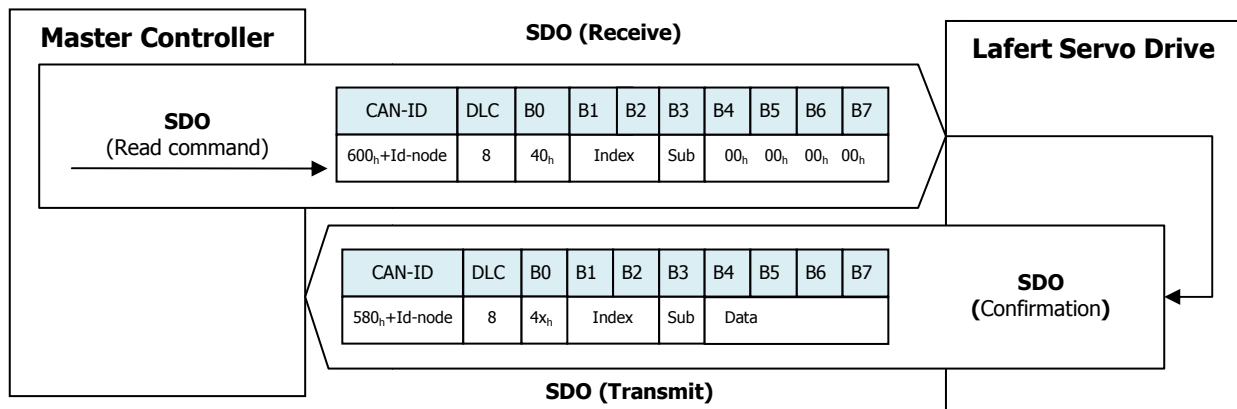
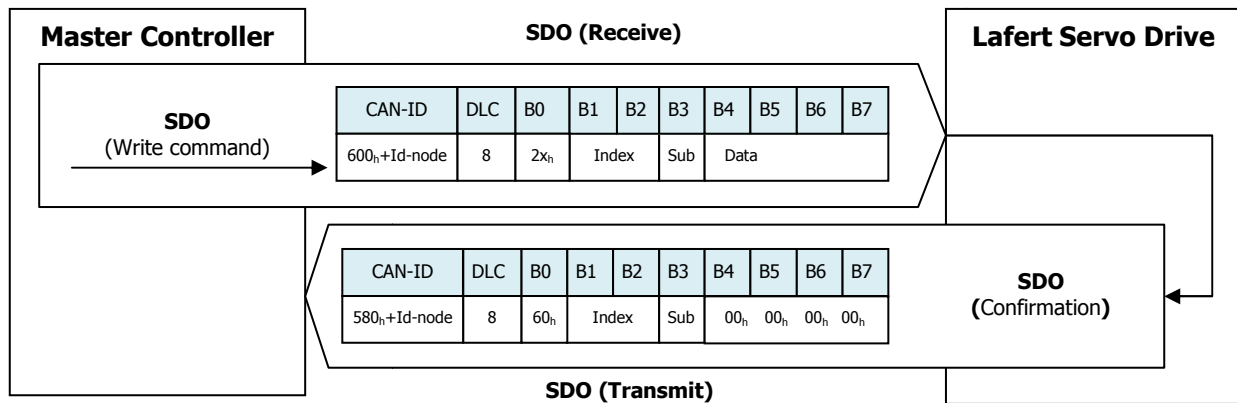
Access via data objects (PDO/SDO)

CANopen makes available a simple and standardised possibility for accessing the parameters of the Lafert Drive (i.e. Target Speed or profile Acceleration). A unique number (index and sub-index) is assigned to each parameter (CANopen object). The totality of all adjustable parameters is contained in the object directory (OD).

All objects are defined in EDS file (electronic Data Sheet).

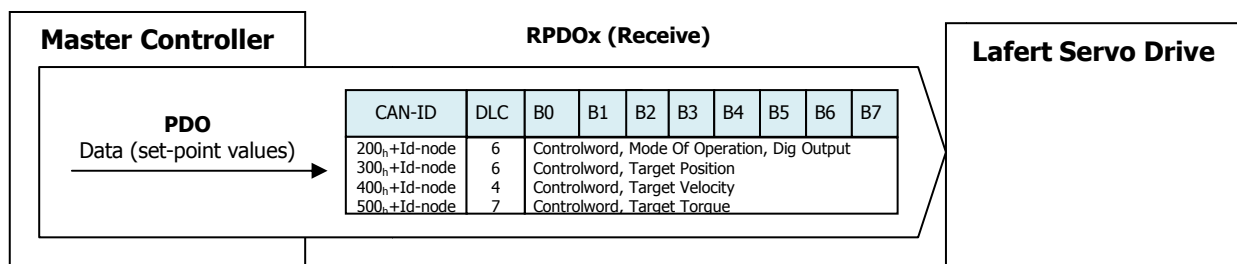
There are 2 methods for accessing CANopen objects via the CAN bus:

- **Access via Service data object (SDO):** confirmed type of access where the Lafert Drive acknowledges every parameter access

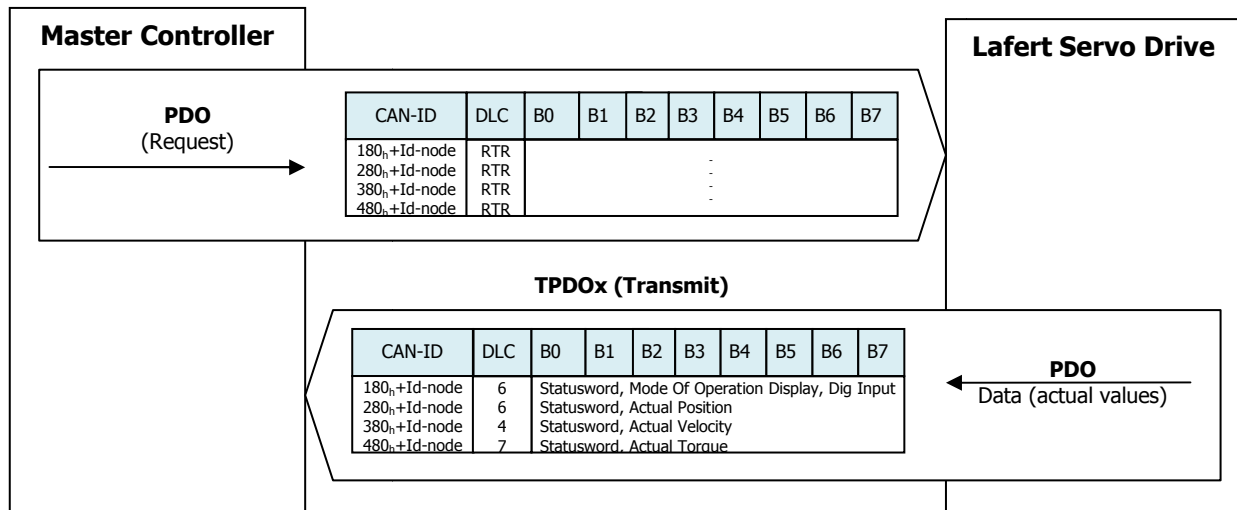


- **Access via Process data objects (PDO):** unconfirmed type of access for which no acknowledgement takes place.

The Receive PDOs are composed by controlword and set points values:

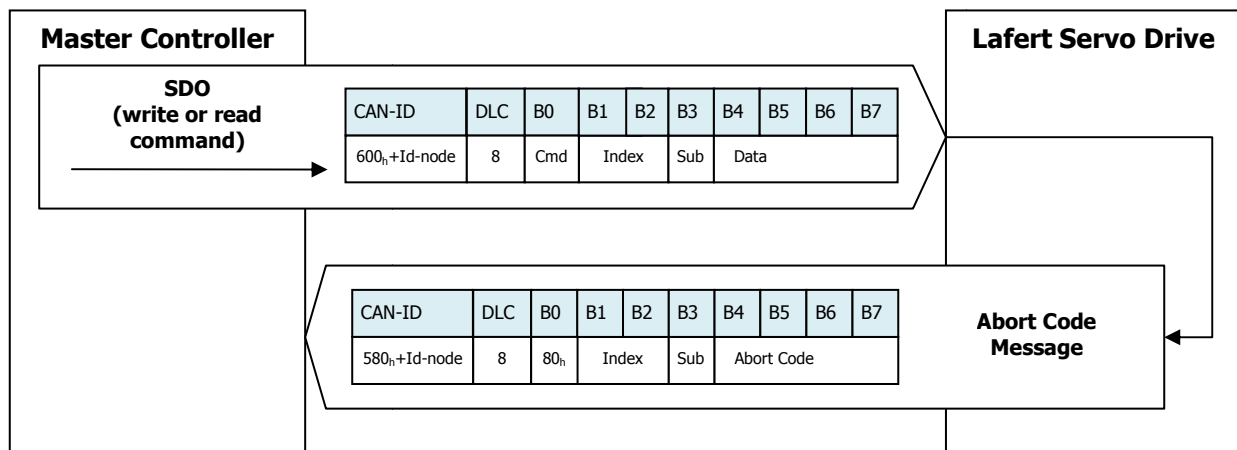


The Transmit PDOs are composed by statusword and actual values:



- **Abort Code via Service data object (SDO):** case of an error when reading or writing (for example, because the written value is too large), the Lafert Servo Drive answers with an error message instead of the acknowledgement.

The cause of the fault was returned to the data (Abort Code) as an error code.

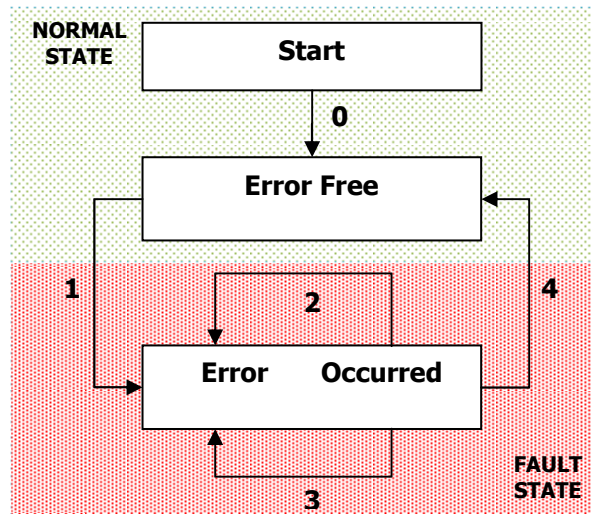


6.6 Emergency Protocol (EMCY)

The Lafert Servo Drive monitors the function of internal modules and of the firmware. Whenever an error occurs, the parameterised error response is initiated and the corresponding EMCY message is transmitted.

The latest error message is always stored here in Error Code object (603F_h:0_h). Also, it is in the highest error memory slot (1003_h: 01_h), the error memory always saves the 15 most recent error messages that can also be read out.

The Lafert Servo Drive also transmits an EMCY message once the error has been acknowledged.

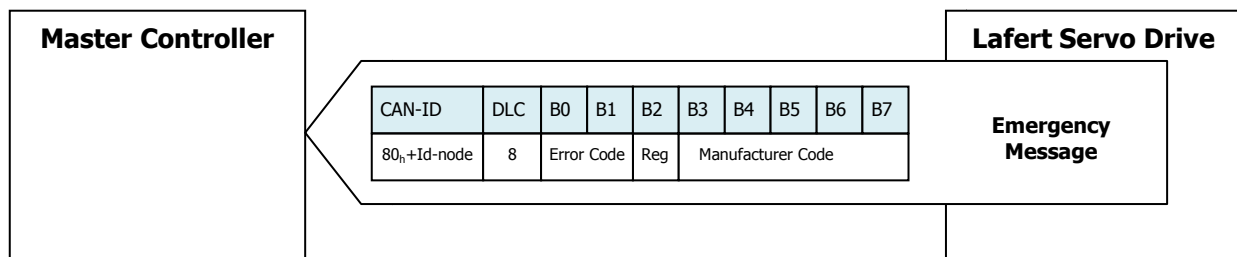


The following status transitions are possible:

Transition	Cause	Description
0	Initialisation completed	There is no error. The drive sends the error code 0000 _h (Error reset/No error) and the state of CAN (8170 _h).
1	Error occurs	No error was present and a new error occurs. The drive goes to Fault State. Verify the diagnostic state and the Emergency message.
2	Error acknowledgment not successful	Not all causes of error have been remedied and an error acknowledgement was performed.
3	New error occurs	There is an error and a new error occurs. An EMCY message with the error code for the new error (1003 _h : 01 _h , standard error field 1) is written.
4	Error acknowledgment successful	All causes of error have been remedied and an error acknowledgement was performed. The EMCY message was transmitted with error code 0000 _h (Error reset/No error).

Emergency objects

The Lafert Servo Drive transmits an Emergency (EMGY) message whenever an error occurs.



The CAN identifier (CAN-ID) is 80_h + Id-node. The data message identifies one univocal Error Code, it defines the alarm occurred.

For referring the defines of alarm see the "Lafert – CANOpen Manual"

6.7 Heartbeat and Node Guarding

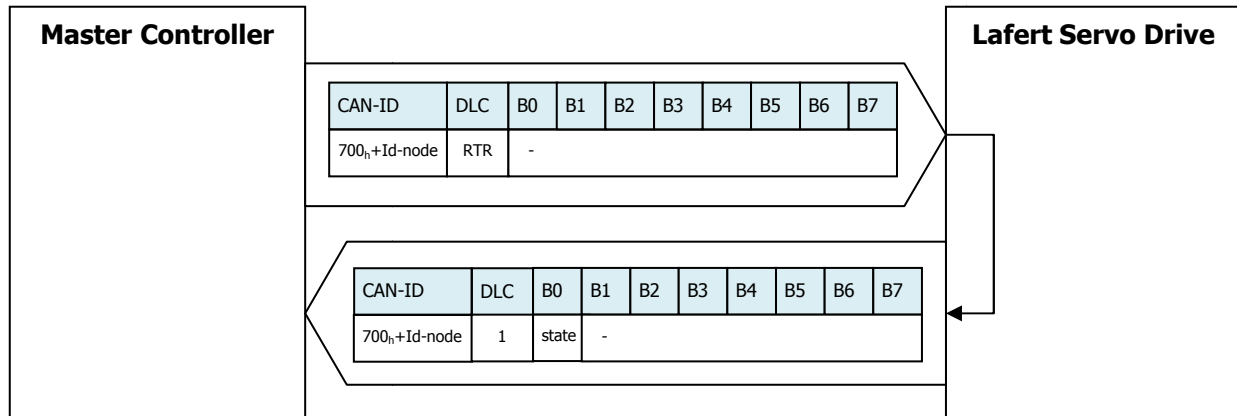
There are two types of error control protocol: Node Guarding Protocol and Heartbeat protocol.



DS301 recommends using heartbeat in the new implementations due to the sensible nature of Guarding using the RTR frames.

Node Guarding Protocol

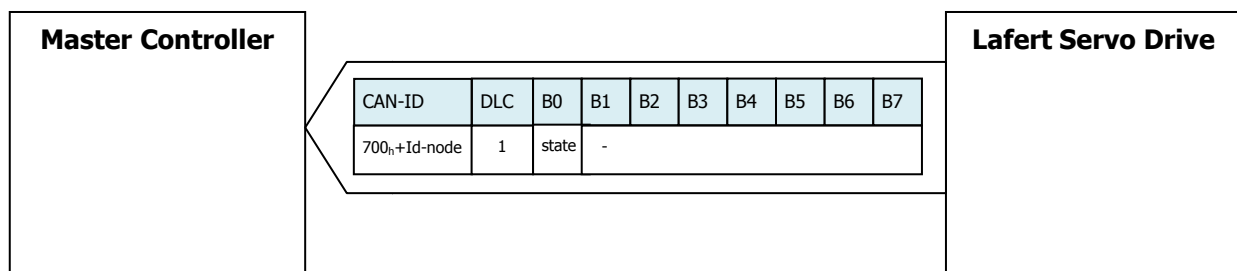
The guarding is achieved through transmitting guarding requests (Node guarding protocol) by the NMT Master. If a NMT Slave has not responded within a defined span of time (node life time) or if the NMT Slave's communication status has changed, the NMT Master informs its NMT Master Application about that event.



Heartbeat Protocol

The Heartbeat protocol is used to monitor the nodes in the network and verify that they are alive.

A heartbeat producer (usually a slave device) periodically sends at node id (700_n+Id-node). The data part of the frame contains a byte indicating the node status.

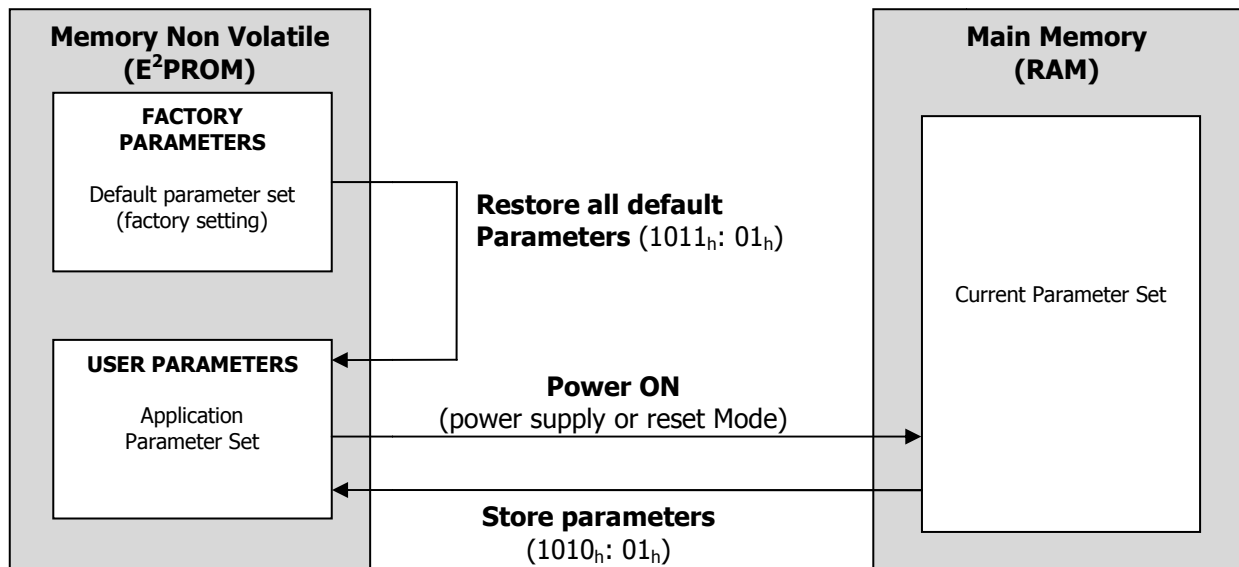


The heartbeat consumer reads these messages. If the messages fail to arrive within a certain time limit (defined in the object directory of the devices) the consumer can take action to, for example, reset the device or indicate an error. Frame format is: COBID + DATA (status of node)

A Heartbeat Producer transmits a Heartbeat message cyclically. One or more Heartbeat Consumer receives the message. The relationship between producer and consumer is configurable via the object dictionary.

6.8 Store and Restore

The parameter sets contain all of the parameters required for operation.



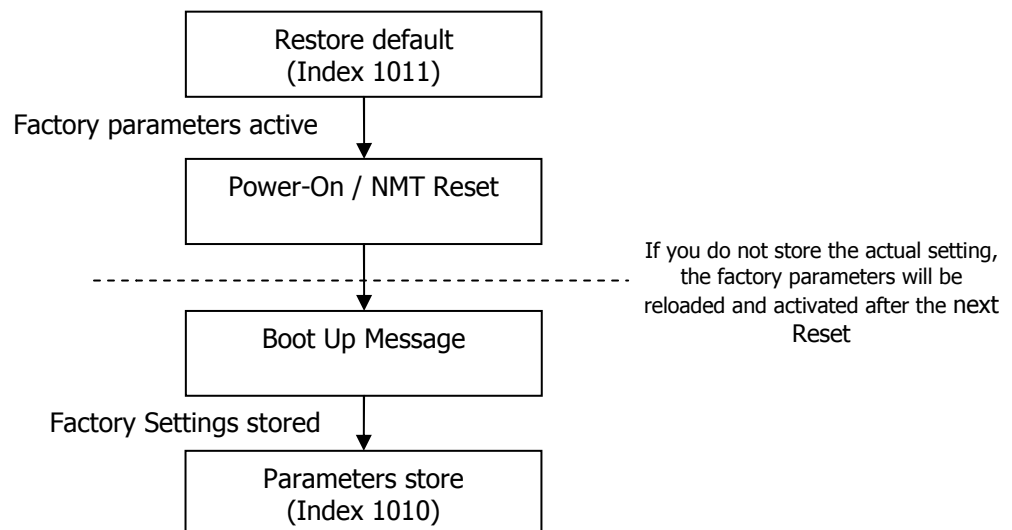
Current parameter set (temporary)

The current parameter set is located in the main memory (RAM) of the Lafert Servo Drive. Via SDO the current parameter set can be written or read in any desired way.

	Caution If there is an interruption in the power supply or a Reset command all changes to the current parameter set are lost. To save the parameter in the memory non volatile using the "Store Parameter" (1010_h: 00_h) object.
---	--

Default parameter set (Factory Settings)

The default parameter set is the standard parameter set of the Lafert Servo Drive provided by the manufacturer and it is unchangeable. Using the object "Restore all default parameters" (1011_h: 01_h) the default parameter set can be loaded into the current parameter set in main memory.

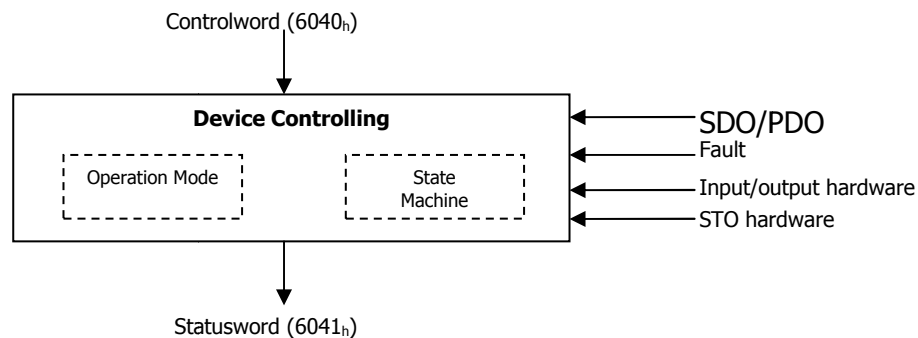


Application parameter set

The application parameter set is loaded into the current parameter set of the main memory automatically at every Power ON of the power supply. Using the object "Store parameters" (1010_h: 01_h) the current parameter set can be saved into the application parameter set in permanent memory.

6.9 Device Control

The Lafert Servo Drive is controlled by CANopen using the Finite State Machine by profile DSP402. State may be changed using the *controlword* (6040_h) and/or according to internal events. The current state can be read using the *statusword* (6041_h).

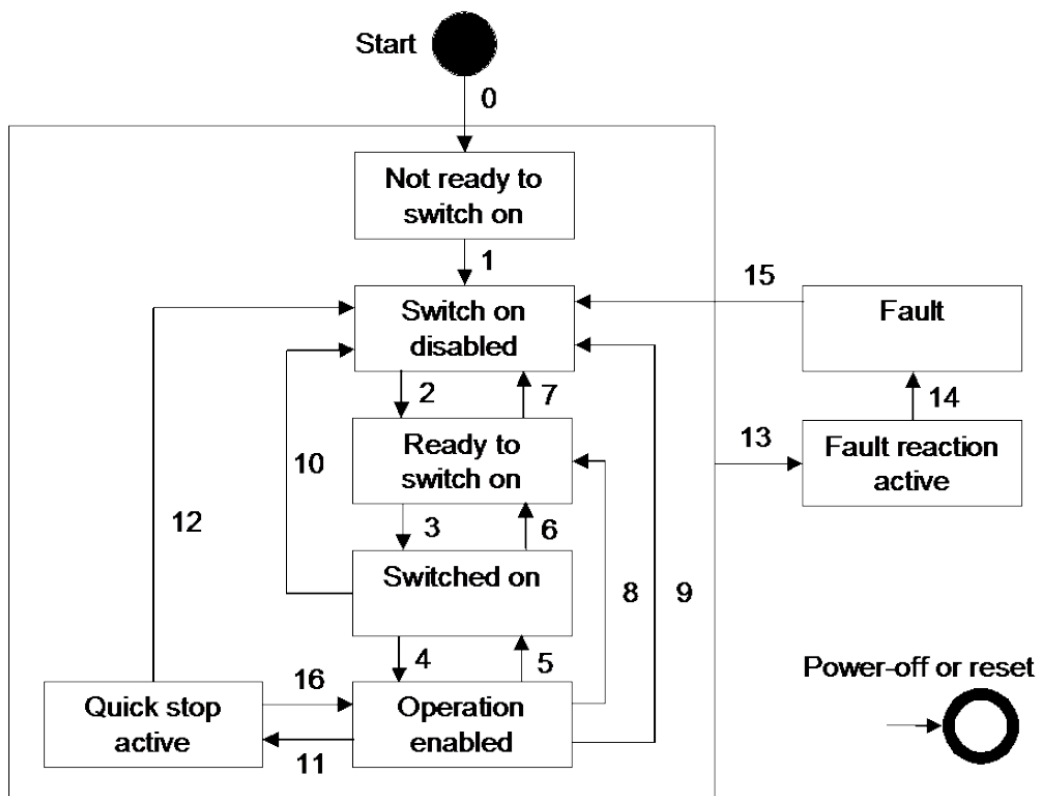


Finite State Machine by DSP402

The device control is performed by a state machine according to DSP402.

State changes are triggered by internal events such as the occurrence of an error or external demand by means of *Controlword* (6040_h). *Statusword* (6041_h) gives feedback about the actual state.

After power-up and initialisation, the drive switches to the state "**Switch On Disabled**" automatically. In the state "**Operation Enabled**", the drive is fully operational.



State	Description
START	The status is left after Power ON or at a Reset node and it changes automatically into "ready to switch on" mode. An existing brake is closed.
NOT READY TO SWITCH ON	The CAN bus interface is initialised and the CAN bus communication is approved. The output stage is switched off An existing brake is closed.
SWITCH ON DISABLED	The CAN bus communication is enabled and from this status, parameters can be updated. Status transitions can only be controlled by commands or triggered by error events. The output stage is switched off. An existing brake is closed, or gets closed.
READY TO SWITCH ON	The output stage is switched off. An existing brake is closed, or gets closed.
SWITCHED ON	The output stage is switched on. An existing brake is opened, or gets opened.
OPERATION ENABLE	The drive is controlled in accordance with its parameterized operating mode (e. g. velocity task or set-point values). The output stage is switched on. An existing brake is opened.
QUICK STOP ACTIVE	The output stage is switched on. An existing brake is opened. The drive stops the motion and either stays in quick stop with torque applied.
FAULT REACTION ACTIVE	An error has occurred. An existing brake is opened.

FAULT	An error has occurred. The drive remains in fault condition, until receives a Reset Fault.
--------------	---

The drive moves in the state SAFETY from all states. To exit by Safety State it is necessary to send the controlword with command "Disable Voltage".

Object parameters of device control

The following objects are available for the control state machine.

Index	Object	Name	Type	M/O	Attr.	Description
6040_h	VAR	controlword	UINT16	M	rw	This object shall indicate the received command controlling the PDS FSA.
6041_h	VAR	statusword	UINT16	M	ro	This object shall provide the status of the PDS FSA.
6060_h	VAR	mode of operation	INT8	M	rw	This object shall indicate the requested operation mode
6041_h	VAR	mode of operation display	INT8	M	ro	This object shall provide the actual operation mode.
603F_h	VAR	Error Code	U32	M	ro	This object shall provide the error code of the last error which occurred in the drive device.
60FD_h	VAR	Digital Inputs	U32	M	ro	This object shall provide digital inputs state.
60FE_h	VAR	Digital Outputs	U32	M	rw	This object shall command the digital outputs.

For object's management referring to "Lafert – CANOpen Manual" document.

6.10 Profile Mode Supported

Different operation modes are available with the CiA 402 profile.

The "Supported Drive Mode" object (6502_h) defines the profiles mode supported.

It is possible to change the profile using the "Mode of Operation" object (6060_h) for following profile:

- **Mode (1) Profile position mode:** *it is not available yet.*
- **Mode (3) Profile velocity mode:** Reference velocity assignment by controller.
- **Mode (4) Profile torque mode:** *it is not available yet.*
- **Mode (6) Homing Mode:** *it is not available yet.*

The following profiles are supported but for using contact the manufacturer.

- **Mode (-1) Analogue Mode:** the drive is controlled by serial command or hardware digital input
- **Mode (-2) Profile Reserved**

6.11 Profile Velocity

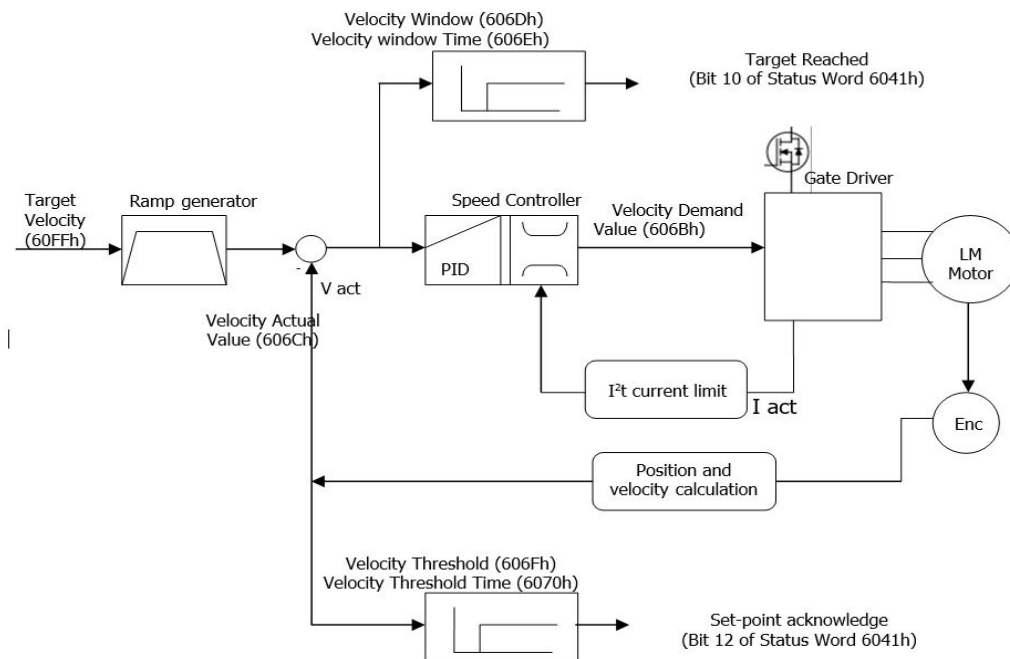
This mode operates the motor in Velocity Mode with trapezoidal case. In the Profile Velocity operation mode the movement profile is defined by velocity and acceleration/decelerations commands. To activate the mode, the vale "3" must be set in "mode of operation" object (6060_h).

Object entries

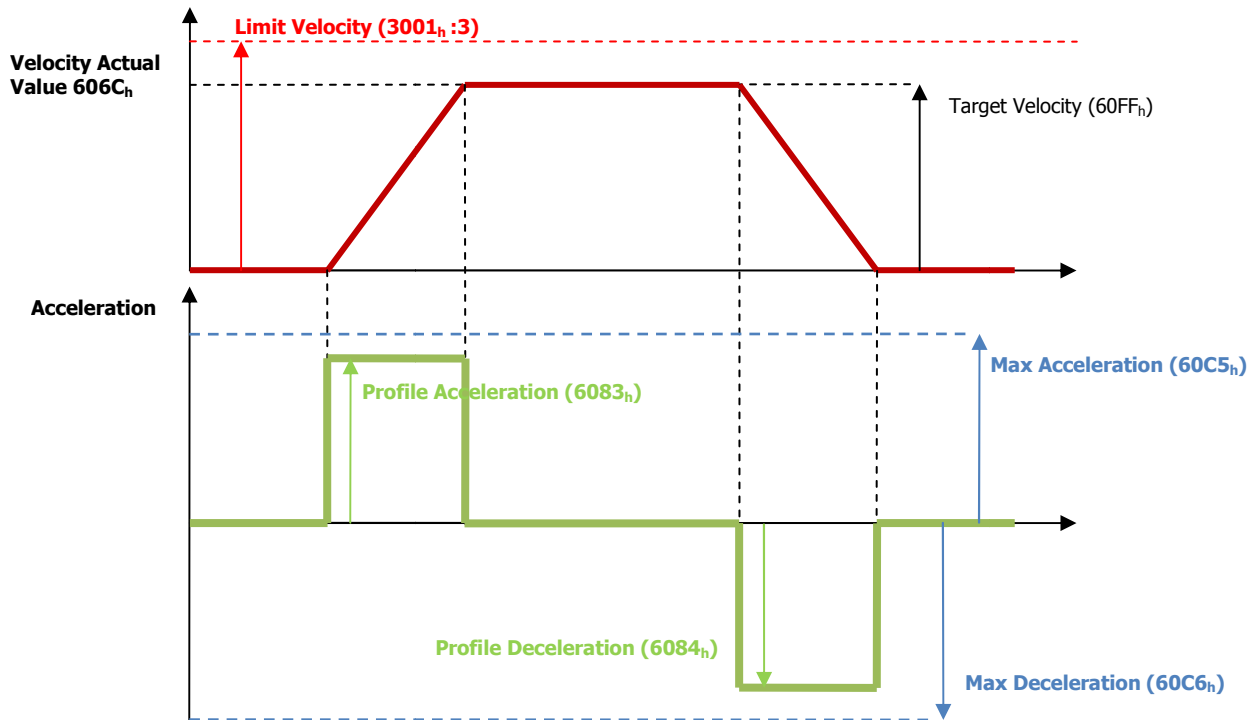
The following objects are available for the control profile velocity and the behaviour of the drive.

Index	Object	Name	Type	M/O	Attr	Description
60FF _h	VAR	Target Velocity	INT32	M	rw	This object shall indicate the configured target velocity and shall be used as input for the trajectory generator. The value shall be given in user-defined velocity units.
607F _h	VAR	Max Profile Velocity	UINT32	O	rw	This object shall indicate the configured maximal allowed velocity in either direction during a profiled motion.
6083 _h	VAR	Profile Acceleration	UINT32	O	rw	This object shall indicate the configured acceleration. The value shall be given in user-defined acceleration units.
6084 _h	VAR	Profile Deceleration	UINT32	O	rw	This object shall indicate the configured deceleration. The value shall be given in user-defined acceleration units.
607E _h	VAR	Polarity	UINT8	O	rw	This object shall indicate if the position demand value shall be multiplied by 1 or by -1.
606B _h	VAR	Velocity Demand Value	INT32	O	ro	This object shall provide the output value of the trajectory generator. The value shall be given in the user-defined velocity units.
606C _h	VAR	Velocity Actual Value	INT32	M	ro	This object shall provide the actual velocity value derived either from the velocity sensor or the position sensor. The value shall be given in user-defined velocity units.
606D _h	VAR	Velocity Window	UINT16	O	rw	This object shall indicate the configured velocity window. The value shall be given in userdefined velocity units.
606E _h	VAR	Velocity Window Time	UINT16	O	rw	This object shall indicate the configured velocity window time. The value shall be given in milliseconds.
606F _h	VAR	Velocity Threshold	UINT16	O	rw	This object shall indicate the configured velocity threshold. The value shall be given in userdefined velocity units
6070 _h	VAR	Velocity Threshold Time	UINT16	O	rw	This object shall indicate the configured velocity threshold time. The value shall be given in milliseconds.

Controller structure



Diagrams Profile Velocity Mode



6.11.1 RUN Sequence

- 1) Switch ON Power Supply
- 2) Verify LED Status 1/2 in INIT Mode
- 3) Configure Profile Velocity 0x6060 → 0x03
- 4) Set **READY TO SWITCH ON** State: write Control Word 0x6040 → 0x06
- 5) Set **SWITCHED ON** State: write Control Word 0x6040 → 0x07
- 6) Verify that the Smartris Drive is in SWITCHED ON: read Status Word 0x6041 → 0x23
- 7) Set **OPERATION ENABLED** State: write Control Word 0x6040 → 0x0F
- 8) Verify LED STATUS ENABLED
Verify that the Smartris Drive is in ENABLED: read Status Word 0x6041 → 0x27
Verify that the brake is released
- 9) Write Speed Set Point: 0x60FF → 0x03e8 (for example set 1000 rpm)
- 10) Verify if the motor is running
Verify the motor speed (after ramp): read 0x606C → 0x03e8 (for example set 1000 rpm)

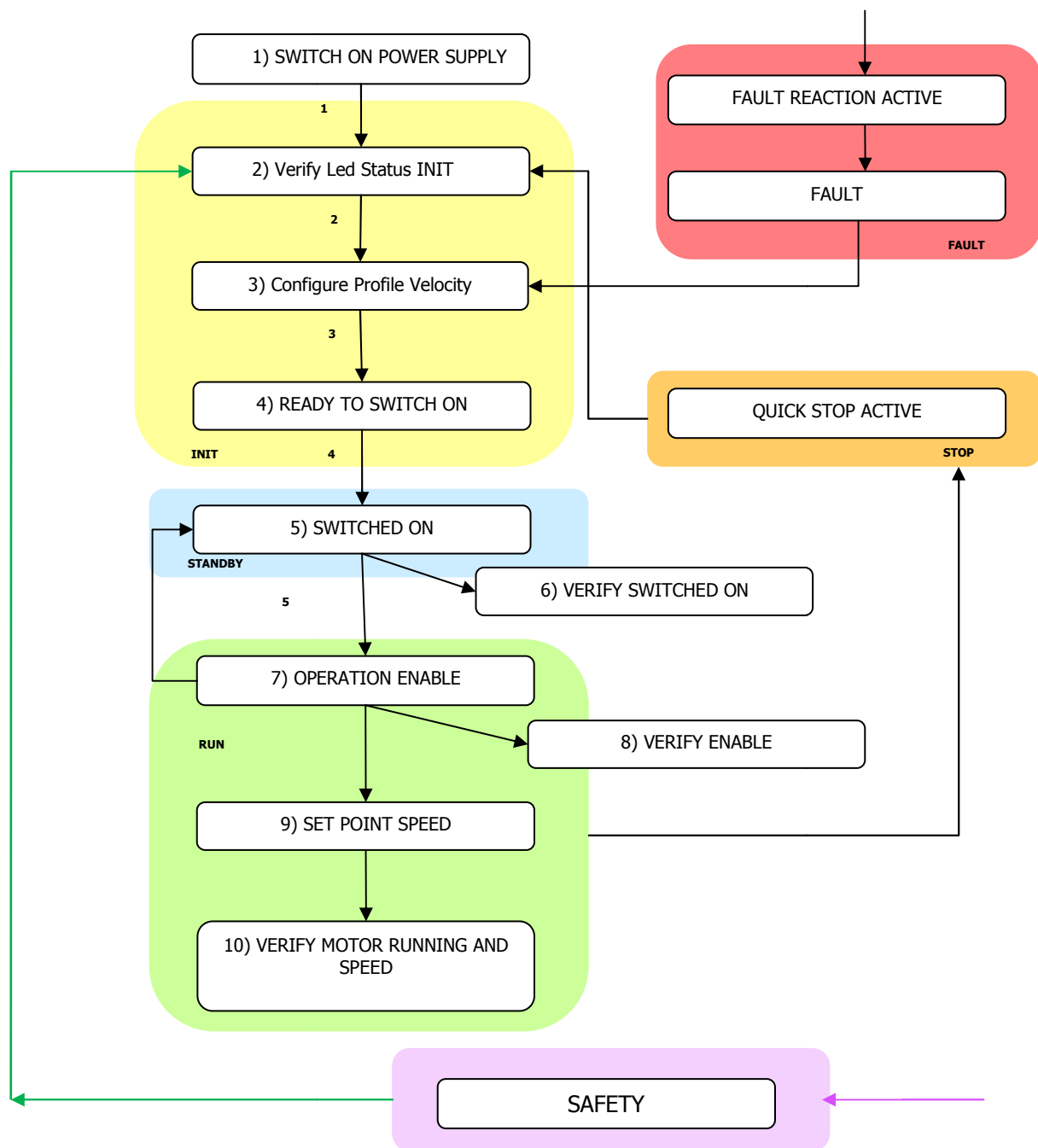
NOTES:

- The STO (**SAFETY**) command may stop the running command immediately
- The **Stop** Command can stop the running command immediately

- A **FAULT** (see table in Diagnostic) can stop the running command immediately



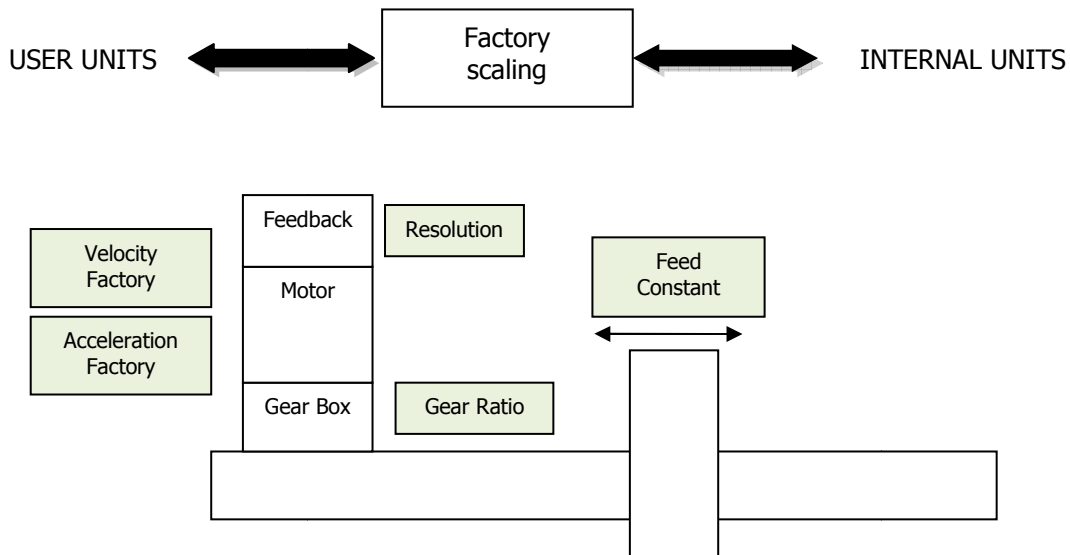
See picture below to the flow chart of running sequence



Picture 14 - CANOpen Run Sequence

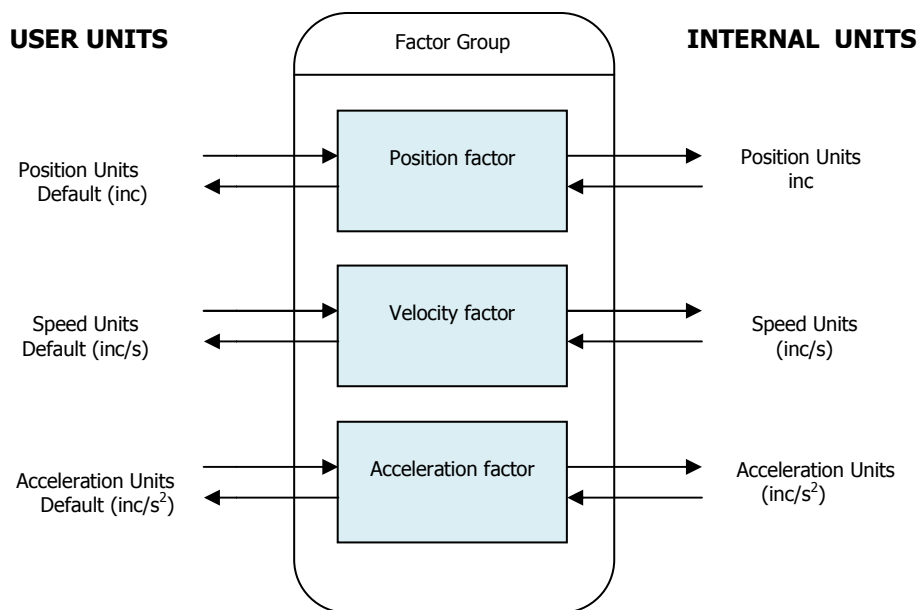
7 | MEASURING UNIT CONVERSION

The drive can be used in different applications. Internal values are entered and calculated in increments, they are dependent on the resolution of the feedback used. The user parameters are defined in increments as default. For setting parameters easily in different applications it is possible to convert the user unit in other units.



Picture 15 - Factory group

It defines [u.u.] as user unit and [i.u.] as internal units.



Picture 16 - Factory group units

The values of the Velocity Factory Group and Acceleration Factory group are used to convert internal values of speed and acceleration parameters into user-defined units.

The Velocity Factor group is composed by numerator and divisor and it has to be entered separately.

$$\text{Velocity Factor} = \frac{\text{Numerator}}{\text{Divisor}}$$

The default value of user unit of velocity is [inc/s]: the numerator and the divisor are set "1".

$$\text{Velocity}[i.u.] = \text{Velocity}[u.u.] \times \left(\frac{60}{\text{Resolution}} \right) \times \left(\frac{\text{Numerator}}{\text{Divisor}} \right)$$

The Acceleration Factor is composed by numerator and divisor and it has to be entered separately.

$$\text{Acceleration Factor} = \frac{\text{Numerator}}{\text{Divisor}}$$

The default value of user unit of acceleration is [inc/s²]: the numerator and the divisor are set "1".

$$\text{Acceleration}[i.u.] = \text{Acceleration}[u.u.] \times \left(\frac{60}{\text{Resolution}} \right) \times \left(\frac{\text{Numerator}}{\text{Divisor}} \right)$$



Warning

The drive has as a default unit the increments. If you want to change the units see the procedure. After the units changing It is mandatory modify also some objects (for example profile acceleration/deceleration, max acceleration/deceleration ...)

Measuring Unit Conversion Parameters

In CANopen it is possible to change the user unit group using these objects

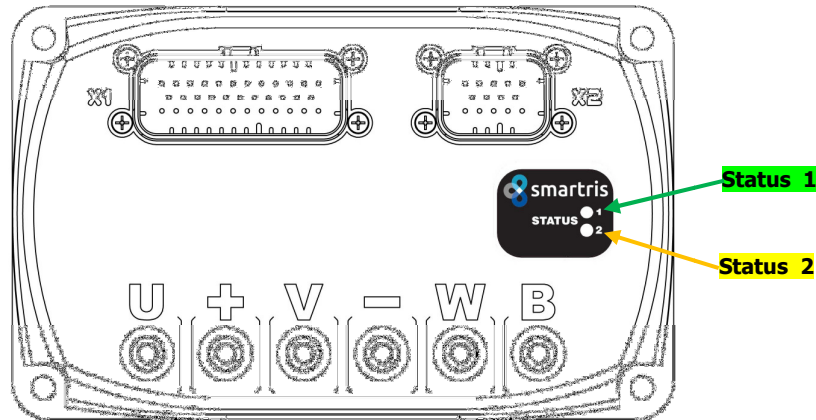
Index	Object	Name	Type	M/O	Attr.	Description
6096 _n	VAR	Velocity Factor	UINT32	M	rw	This object can be used to match the velocity units to the user-defined velocity units
6097 _n	VAR	Acceleration Factor	UINT32	M	rw	This object can be used to match the acceleration units to the user-defined acceleration units.

For object's management referring to "Lafert – CANOpen Manual" document.

Table Conversion

FEEDBACK	Velocity Factor		Velocity	Acceleration Factor		Acceleration
Resolution	NUM	DIV	[u.u]	NUM	DIV	[u.u]
4096	1	1	[inc/s]	1	1	[inc/s ²]
	4096	60	[RPM]	4096	60	[RPM/s]
16384	1	1	[inc/s]	1	1	[inc/s ²]
	16384	60	[RPM]	16384	60	[RPM/s]

8 | DIAGNOSTIC



There are 2 leds on right of the top:

- the **GREEN Led** (Status 1) is on upper side and shows the status 1 of the drive
- the **YELLOW Led** (Status 2) is on down side and shows the status 2 of the drive

MACRO DRIVE STATE	CANOpen STATE	STATUS 1 LED GREEN	STATUS 2 LED YELLOW	LED VIEW
INIT	Not Ready To Switch On	"BLINK" simultaneously	"BLINK" simultaneously	 1 simultaneously  2 simultaneously
	Switch On Disabled Ready to Switch On	"BLINK" alternately	"BLINK" alternately	 1 alternately  2 alternately
STANDBY	Switched On	"BLINK"	OFF	 1 BLINK 50%  2 OFF
FAULT	Fault Fault reaction fault	"BLINK" [x]	"BLINK" [y]	 1 see fault  2 chapter
RUN	Operation Enabled	ON	OFF	 1 ON  2 OFF
STOP	Quick Stop Active	ON	ON	 1 ON  2 ON
SAFETY	-	OFF	"BLINK"	 1 OFF  2 BLINK

Table 16 - LED Status - Diagnostic

To connect drive to laptop by RS232 and read diagnostic or parameter with Lafert Tools USB to RS232 interface is needed and one suggested is:

- Chipi-X10 Cables (FTDI Manufacturer)

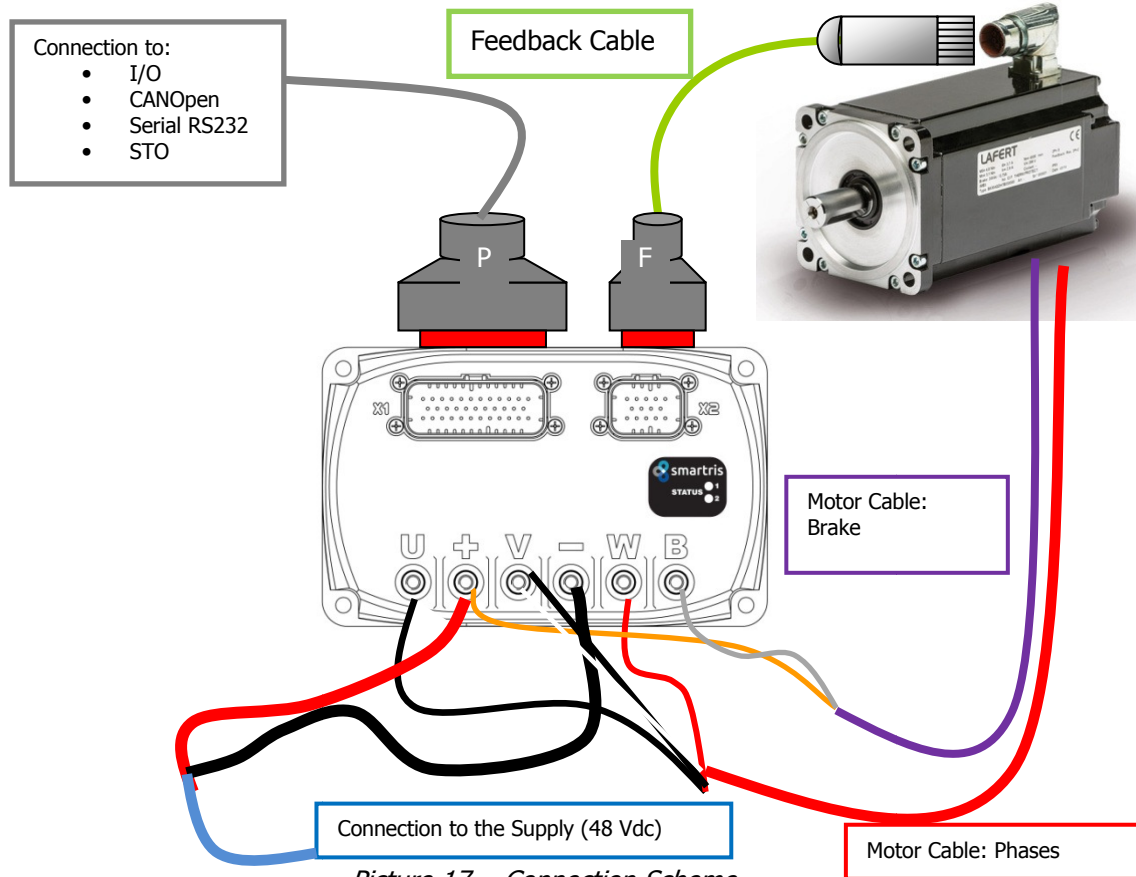
<https://www.ftdichip.com/Products/Cables/USBRS232.htm>



Alarm	STATUS 1 CODE LED GREEN	STATUS 2 CODE LED YELLOW	Alarm Description
	 1st Code	 2nd Code	
A Group: (Temperature)			
Motor Over Temperature	1	10	Motor Temperature over threshold. The motor has reached a too high temperature for correct operation.
Heat Sink Over Temperature	1	1	Heat Sink Temperature over threshold. The Heat Sink has reached a too high temperature for correct operation.
Heat Sink Temperature Out Of Range	1	3	Heat Sink Temperature Sensor is out of range. Potential malfunction of the temperature sensor. (Contact the Manufacturer).
Board Over Temperature	1	4	Internal Board Temperature over threshold. Too high a temperature for correct operation inside the drive.
Board Temperature Out Of Range	1	5	Internal Temperature Sensor out of range. Potential malfunction of the temperature sensor. (Contact the Manufacturer).
Motor Temperature Out Of Range	1	6	Motor Temperature Sensor is out of range. Potential malfunction of the temperature sensor. (Contact the Manufacturer).
B Group: (Feedback)			
Resolver	2	10	Check resolver connections, connectors and wiring of both sides.
Resolver Initialization	2	4	Initialization Fault for Resolver Device. (Contact the Manufacturer).
Encoder	2	5	Incremental Encoder Fault
SinCos Fault	2	6	SinCos Encoder Fault
Hall	2	7	Hall Sensors Fault
Distance Hall	2	8	Hall Sensors Fault
C Group: (Current)			
Offset Current Sensor	3	10	Offset current sensor is out of range. (Contact the Manufacturer).
Over Current	3	1	The current absorbed by the motor is beyond the set limit. Check Phase Motor connection and wire. Look for any short circuits.
D Group: (Voltage)			
Under Voltage	4	1	DC Bus voltage value lower than the limit threshold. Check mains voltage at terminals +, -.
Over Voltage	4	2	DC Bus voltage value higher than the limit threshold. Check mains voltage at terminals +, -
E Group: (Functionality)			
Velocity Fault	5	10	The actual speed differs from the target Speed.
I ² T Overload Protection	5	2	I2T overload motor protection reached.
Hardware	5	3	Error Hardware (Contact the Manufacturer)
External HW	5	4	Error CAN Interface (Contact the Manufacturer)
F Group: (Communication)			
E ² prom	6	1	Parameter Fault stored in E2prom.
CanOpen	6	2	Communication Fault with CANOpen
Sincos Fault	6	3	Internal Communication Fault (Contact the Manufacturer).
Configuration Parameters	6	4	Configuration Parameters Fault (Contact the Manufacturer).
Profile Generic	6	5	Error Configuration Profile: Mode Of Operation
Torque Profile	6	6	Error Torque Profile
Velocity Profile	6	7	Error Velocity Profile
Configuration Parameters			
Program Fault	7	X	Code Programming Fault (Contact the Manufacturer).
	8	X	
	9	X	

Table 17 - Error Code

9 | CONNECTION SCHEME



Picture 17 - Connection Scheme

For the connection of the different cables see the tables on Chapter 1:

POWER SUPPLY SMARTRIS WIRING			
PIN	SIGNAL	DESCRIPTION	COLOR (EXAMPLE)
+	+ VBUS	Connection to + Vbus (+48 Vdc)	Red
-	- VBUS	Connection to - Vbus	Black

Table 18 - Power Supply Wiring Scheme

MOTOR PHASES SMARTRIS WIRING			
PIN	SIGNAL	DESCRIPTION	COLOR
U	PHASE U	Motor Phase U	Black
V	PHASE V	Motor Phase V	White
W	PHASE W	Motor Phase W	Red

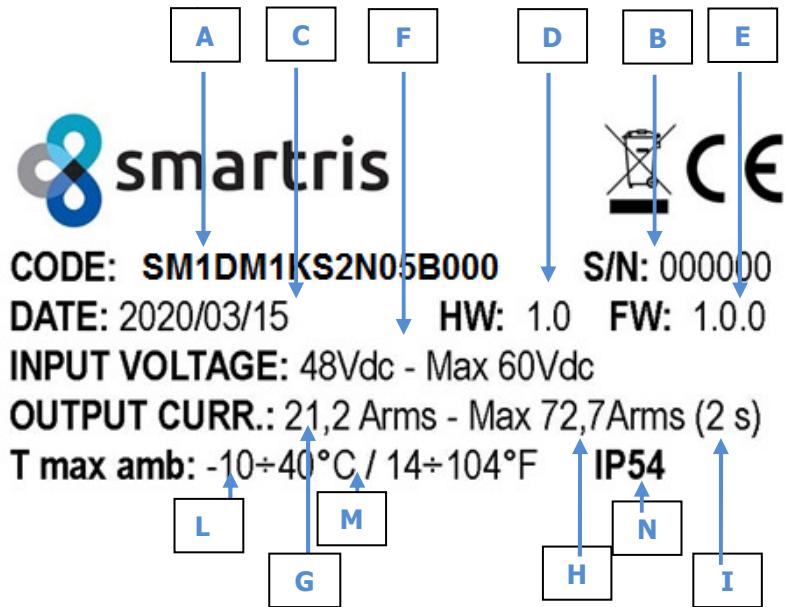
Table 19 - Motor Phases Scheme

BRAKE			
PIN	SIGNAL	DESCRIPTION	COLOR (EXAMPLE)
+	+BRAKE	Motor Brake + (connect to + Vbus)	Orange
B	-BRAKE	Motor Brake -	Grey

Table 20 - Brake Scheme (see Wiring Example Chapter 1)

9.1 Label/Code

Label

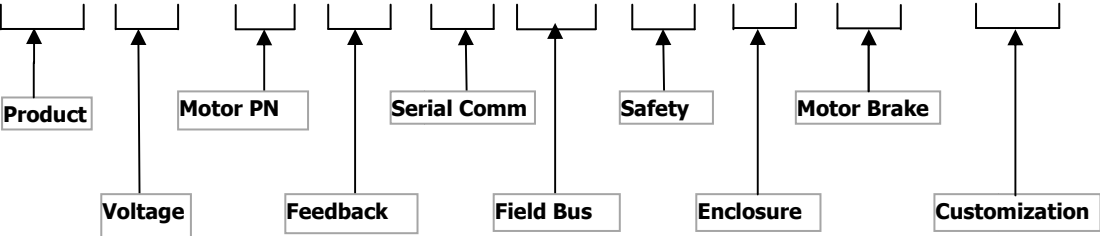


Mark	Description	Mark	Description
A	Code	G	Rated Current
B	Serial Number	H	Peak Current
C	Production Date	I	I2T Time
D	HW Release	L	Minim Ambient Temperature
E	FW Release	M	Maximum Ambient Temperature
F	Voltage	N	Protection Class

Table 21 - Label Description

Using the following code, the properties of the Smartris can be determined (the code is an example):

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Product	Voltage	Motor	Feedback	Serial	Field Bus	Safety	Enclosure	Motor Brake	Customization						
SM1	G	XX	05	2	C	1	5	B	000						
Smartris	48V	Custom	Resolver 2 poles	RS232	CANOpen	STO	IP54	Brake Present	No custom						



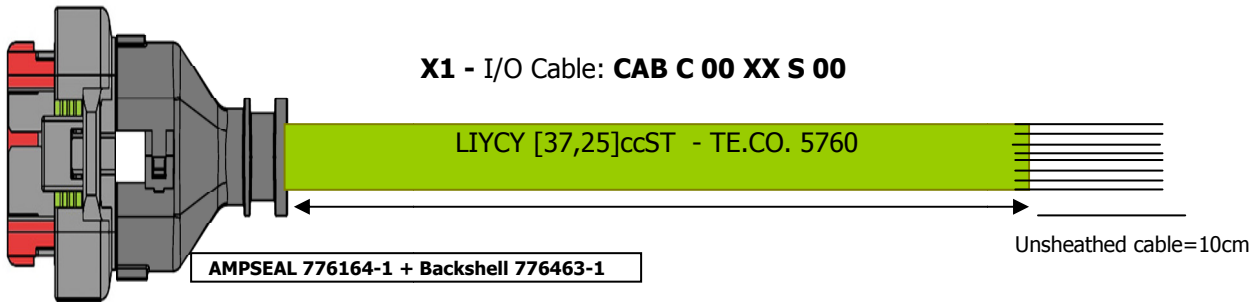
Picture 18 - Code Example

PRODUCT	MOTOR PN	SERIAL COMM	SAFETY	MOTOR BRAKE
3 Letters	2 Letters	1 Letters	1 Letters	1 Letters
Pos 1-3	Pos 5-6	Pos 9	Pos 11	Pos 13
SM1 - Smartris I Gen	S1 - B40 / 508E-21 / 4500	2 - RS232	0 - NO STO	B - Brake present
SMT - Solo Drive	M1 - B63 / 509E-21 / 4500	4 - RS485	1 - STO	S - Spring Brake
	L1 - B71 / 510R-26 / 3000			N - No brake
	XX - Custom			
VOLTAGE	FEEDBACK	FIELD BUS	ENCLOSURE	CUSTOMIZATION
1 Letters	2 Letters	1 Letters	1 Letters	4 Letters
Pos 4	Pos 7-8	Pos 10	Pos 12	Pos 14-16
C - 24 V	05 - Resolver	C- CANOpen DS402	5- IP54	000
D - 36 V	RS - SRS50 N9	E - EtherCAT	2 - IP30	
D - 48V	RM - SRM50 H9	N - No fieldbus	0 - No Enclosure	
	EK - SEK37			
	EL - SEL37			
	KS - SKS36/SKS36S			
	KM - SKM36			
	LE - EKM36			
	LF - EKM36			
	E9 - INC.1000 Imp/Gear			
	09 - INC.1024 Imp/Gear			
	L9 - INC.2000 Imp/Gear			
	F9 - INC.2048 Imp/Gear			
	N9 - INC.2500 Imp/Gear			
	H9 - INC.4096 Imp/Giro			

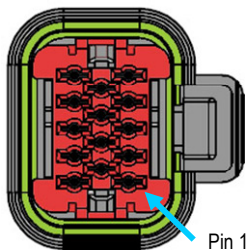
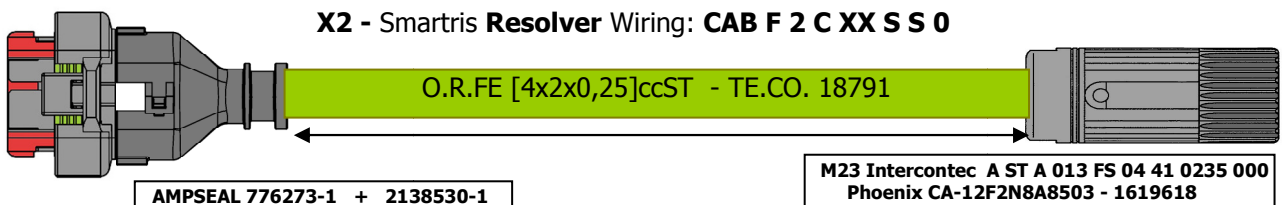
Table 22 - Code Description

APPENDIX A | OPTIONS

Cables for Lafert Standard Motors: XX in code mains the length in m (i.e. XX = 01 means 1 m)



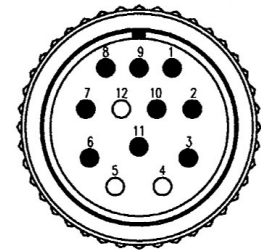
Pin	Colour	Pin	Colour	Pin	Colour
1	Black	13	Brown - Green	25	Blu
2	Pink	14	Brown	26	White -Blue
3	White - Pink	15	Yellow	27	Grey - Blue
4	Grey - Pink	16	Purple	28	Green - Blue
5	Pink - Green	17	White - Yellow	29	Yellow -Blue
6	Brown - Pink	18	White - Black	30	Green
7	Grey- Brown	19	Brown - Black	31	White
8	Grey- Green	20	White - Green	32	N.C
9	Grey- White	21	N.C.	33	N.C
10	Grey	22	Brown - Red	34	Green - Red
11	Red-White	23	Yellow - Brown	35	Yellow- Black
12	Red- Yellow	24	Red	Shield	Cut external sheath.



Contact Side View

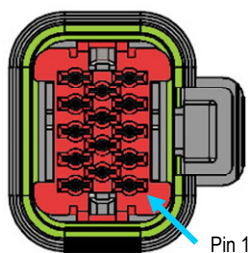
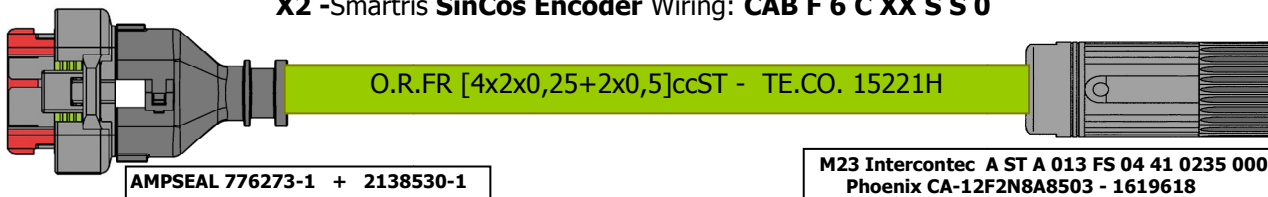
776273-1 14 poles	Description	Color	M23 12 poles
1	S2	White	1
2	S1	Green	2
7	S3	Yellow	3
	NC		4
	NC		5
6	S4	Brown	6
3	R1	Red	7
5	Shield		8
12	NTC1a	Pink	9
13	NTC1b	Grey	10
8	R2	Blue	11
	NC		12

Twisted Pair: (White-Brown), (Green-Yellow), (Grey-Pink), (Blue-Red)



Contact Side View

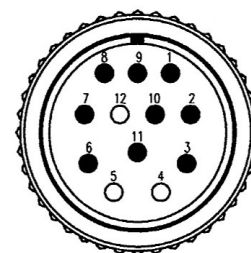
X2 -Smartris SinCos Encoder Wiring: CAB F 6 C XX S S 0



Contact Side View

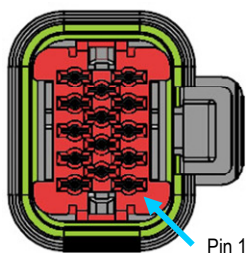
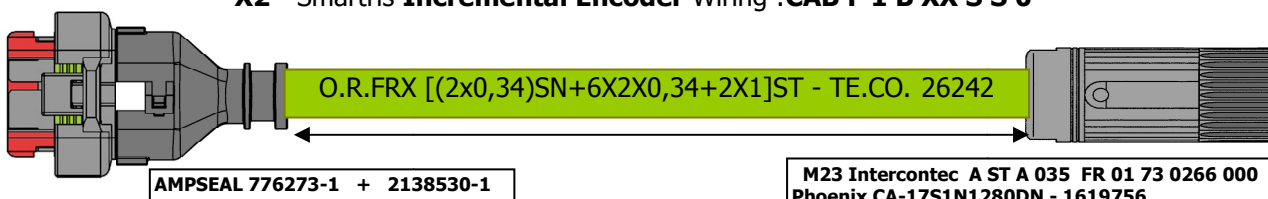
776273-1 14 poles	Description	Color	M23 12 poles
1	+ SIN	Brown	7
6	REF SIN	Green	3
2	+ COS	Blue	8
7	REF COS	Violet	4
3	DATA+	Red	5
8	DATA-	Black	6
10	+V	Brown (0.5mm ²)	1
11	GND	White (0.5mm ²)	2
12	NTC1a	Pink	9
13	NTC1b	Grey	10
5	SHIELD		N.C.
1	+ SIN	Brown	7

Twisted Pair: (Green-Brown), (Blue-Violet), (Red-Black), (Grey-Pink), (Brown 0,5 – White 0,5)



Contact Side View

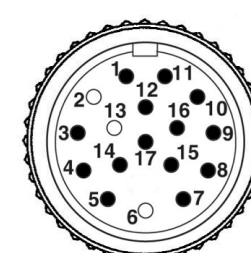
X2 - Smartris Incremental Encoder Wiring :CAB F 1 B XX S S 0



Contact Side View

776273-1 14 poles	Description	Color	M23 17 poles
1	A+	Yellow	9
6	A-	Green	10
2	B+	Red	7
7	B-	Blue	8
3	Z+	Violet	3
8	Z-	Black	4
10	+V	Red (1mm ²)	11
11	GND	Blue (1mm ²)	16
4	HX	Pink	5
9	HY	White-Green	1
14	HZ	Gray-Pink	12
12	NTC1a	White	14
13	NTC1b	Brown	15
15	SHIELD		17

Twisted Pair: (Green-Yellow), (Pink-Gray), (Red-Blue), (Violet-Black), (White/Green – Brown/Green), (Gray/Pink – Red/Blue)



Contact Side View



Caution

Contacts not connected have to be closed with **AMPSEAL 770678-1**

APPENDIX B | DRIVE DISPOSAL

WARNING FOR THE CORRECT DISPOSAL OF THE MOTORS AT THE END OF THEIR LIFE TIME



According to the art. 26 of the Italian law 2014, March 14th n. 49

" Implementation of the **DIRECTIVE 2012/19/EU** OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 4 July 2012, on waste electrical and electronic equipment (WEEE)"

The above symbol, applied to the equipment or its packaging, means that at the end of its lifetime the motors need to be disposed separately from the other waste materials.

The users must dispose their unused motors on behalf of the national authorized collection centers which have been approved for the electrical and electronic waste.

The right collection for recycling, treatment and disposal will contribute to avoid any potential negative environmental & healthy effect and will help both the reuse and the recycling of the motor's components/materials.

TABLE OF DISPOSAL CODES FOR ELECTRIC MOTORS

CODE OF THE MOTOR (WASTE) DISPOSED IN ALL ITS PARTS	EWC (European Waste Code)	EWC DESCRIPTION
In the event of disposal of the full drive	16.02.14	Discarded equipment
In the event of disposal of a disassembled drive:		
• Aluminum parts: heatsink	12.01.03	Non-ferrous metal filings and turnings
• Steel parts: Cover	12.01.01	Ferrous metal filings and turnings
• Plastic parts: fan, fan cover, terminal box	12.01.05	Plastics shavings and turnings
• Electrical components: switches, capacitors, starters, terminal board, electronic boards, IGBT modules...	16.02.16	Components removed from discarded equipment

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